

# Tutorial 12: nMDS and PERMANOVA

ENVX2001 – Applied Statistical Methods

Semester 1

## PERMANOVA, SIMPER and nMDS Analysis

We are going to use the same dataset as last tutorial, but do very different analyses.

### Setup

To do these in R, you need to use the vegan package, so first load it and the MASS package:

```
CODE
library(vegan)
library(MASS)
library(ggplot2)
```

### Load the data

Next load the bird data:

```
CODE
Birds ← read.csv("data/macnally.csv", header = TRUE)
```

### Transform the data

Let's transform the data using a fourth-root transformation:

```
CODE
TransBirdsData ← (Birds[, 3:104])^(1/4)
```

### Bray-Curtis Dissimilarity Matrix

Now let's generate a Bray-Curtis dissimilarity matrix on this data and perform a nMDS. Note this data has already been standardised.

CODE

```
Bird.dis <- vegdist(TransBirdsData, method = "bray")
```

## nMDS Ordination

Let's plot the nMDS and label with Habitat Type:

CODE

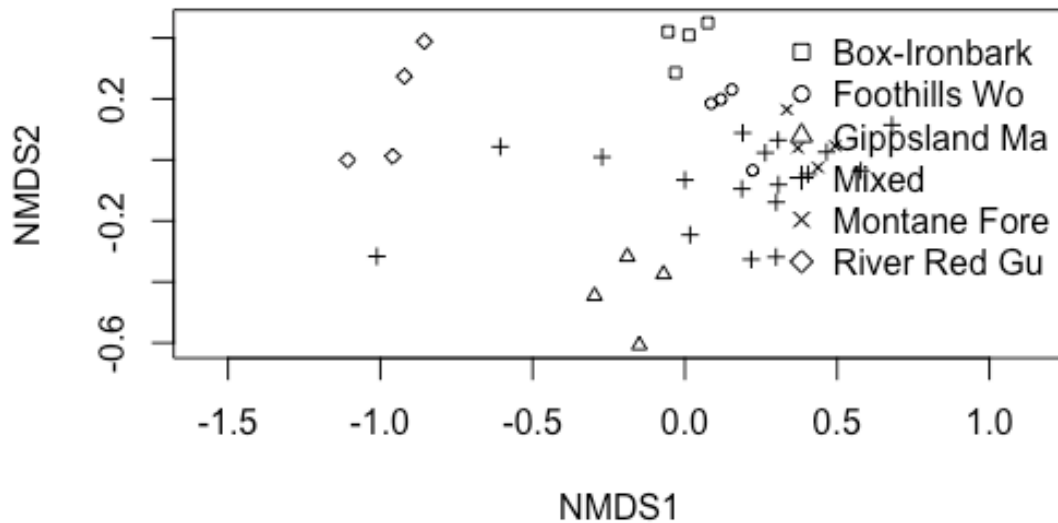
```
pchs <- c(0:5)
Bird_Factor <- factor(Birds$HABITAT)
nMDS_Bird <- metaMDS(TransBirdsData, distance = "bray", k = 2)
```

OUTPUT

```
Run 0 stress 0.1112209
Run 1 stress 0.1368302
Run 2 stress 0.1112209
... Procrustes: rmse 5.373717e-05 max resid 0.0002544127
... Similar to previous best
Run 3 stress 0.129877
Run 4 stress 0.1181205
Run 5 stress 0.111221
... Procrustes: rmse 8.727098e-05 max resid 0.0004126074
... Similar to previous best
Run 6 stress 0.1181205
Run 7 stress 0.1112209
... Procrustes: rmse 7.558244e-06 max resid 3.080781e-05
... Similar to previous best
Run 8 stress 0.129877
Run 9 stress 0.1112209
... Procrustes: rmse 4.061744e-05 max resid 0.0001923771
... Similar to previous best
Run 10 stress 0.1112209
... New best solution
... Procrustes: rmse 8.351695e-07 max resid 3.115428e-06
... Similar to previous best
Run 11 stress 0.1181205
Run 12 stress 0.1268505
Run 13 stress 0.1112209
... Procrustes: rmse 5.560587e-05 max resid 0.0002631007
... Similar to previous best
Run 14 stress 0.1181205
Run 15 stress 0.1389816
Run 16 stress 0.129877
Run 17 stress 0.1112209
... Procrustes: rmse 5.15084e-06 max resid 2.466948e-05
... Similar to previous best
Run 18 stress 0.1112209
... Procrustes: rmse 2.140732e-05 max resid 0.0001003622
... Similar to previous best
Run 19 stress 0.1112209
... Procrustes: rmse 4.55047e-05 max resid 0.000214183
... Similar to previous best
Run 20 stress 0.1112209
... Procrustes: rmse 1.282332e-05 max resid 5.941938e-05
... Similar to previous best
*** Best solution repeated 6 times
```

CODE

```
plot_Bird <- ordiplot(nMDS_Bird, display = "sites", type = "n")
points(nMDS_Bird, col = "black", pch = pchs[Bird_Factor])
legend("topright", bty = "n", legend = levels(Bird_Factor), pch = pchs)
```



CODE

```
nMDS_Bird
```

OUTPUT

```
Call:
metaMDS(comm = TransBirdsData, distance = "bray", k = 2)

global Multidimensional Scaling using monoMDS

Data:      TransBirdsData
Distance: bray

Dimensions: 2
Stress:     0.1112209
Stress type 1, weak ties
Best solution was repeated 6 times in 20 tries
The best solution was from try 10 (random start)
Scaling: centring, PC rotation, halfchange scaling
Species: expanded scores based on 'TransBirdsData'
```

## ANOSIM

Let's do an ANOSIM, using Habitat as the Factor:

**CODE**

```
Bird.anosim <- with(TransBirdsData, anosim(Bird.dis, Birds$HABITAT))
summary(Bird.anosim)
```

**OUTPUT**

```
Call:
anosim(x = Bird.dis, grouping = Birds$HABITAT)
Dissimilarity: bray

ANOSIM statistic R: 0.3936
Significance: 0.002

Permutation: free
Number of permutations: 999

Upper quantiles of permutations (null model):
 90%  95% 97.5%  99%
0.142 0.205 0.269 0.306

Dissimilarity ranks between and within classes:
      0%   25%   50%   75% 100%  N
Between      1 224.25 380.5 520.25 665 500
Box-Ironbark  6  22.50  29.5  45.50   78   6
Foothills Wo 10  47.00  54.0  71.50  145   6
Gippsland Ma 51  72.25  90.0  95.00  130   6
Mixed         4 124.00 220.5 410.25  666 136
Montane Fore  3  17.75  54.5  87.50   97   6
River Red Gu 12  33.75  96.5 179.50  227   6
```

## PERMANOVA

Let's do a PERMANOVA, with HABITAT as a factor. PERMANOVA is called `adonis` in the `vegan` package. Note `adonis` is being replaced by `adonis2`, which is basically the same.

**CODE**

```
adonis2(TransBirdsData ~ HABITAT, data = Birds, permutations = 999)
```

**OUTPUT**

```
Permutation test for adonis under reduced model
Permutation: free
Number of permutations: 999

adonis2(formula = TransBirdsData ~ HABITAT, data = Birds, permutations = 999)
      Df SumOfSqs      R2      F Pr(>F)
Model    5   2.2749 0.50098 6.2245  0.001 ***
Residual 31   2.2660 0.49902
Total    36   4.5409 1.00000
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

## Pairwise Comparisons

If we find an overall significant p-value, we need to see which level is significant from which other level. Here we can do pairwise tests (a bit like post-hoc tests in ANOVAs), with the p-value adjusted with a Bonferroni correction.

To do the pairwise tests, there is a new R command on GitHub. Note the first time you do this, you need to first install Rtools from [here](#).

CODE

```
library(devtools)
install_github("pmartinezarbizu/pairwiseAdonis/pairwiseAdonis")
```

CODE

```
library(pairwiseAdonis)
pairwise.adonis(TransBirdsData, Birds$HABITAT)
```

## SIMPER Analysis

To see what species are contributing to the differences between the vegetation communities, we can use a SIMPER analysis (Similarity Percentages):

CODE

```
Birdsim <- simper(TransBirdsData, Birds$HABITAT)
summary(Birdsim)
```

OUTPUT

Contrast: Mixed\_Gippsland Ma

|         | average  | sd       | ratio    | ava      | avb      | cumsum | p     |     |
|---------|----------|----------|----------|----------|----------|--------|-------|-----|
| V17WPHE | 0.019779 | 0.011557 | 1.711400 | 0.602000 | 2.247700 | 0.040  | 0.001 | *** |
| V34WSW  | 0.018979 | 0.007112 | 2.668500 | 0.247500 | 1.856000 | 0.079  | 0.001 | *** |
| V11CR   | 0.016850 | 0.007496 | 2.247800 | 1.443400 | 0.000000 | 0.114  | 0.001 | *** |
| V6WTR   | 0.015456 | 0.006359 | 2.430500 | 1.325800 | 0.000000 | 0.145  | 0.001 | *** |
| V15STTH | 0.014940 | 0.009012 | 1.657800 | 1.475400 | 0.323400 | 0.176  | 0.003 | **  |
| V25WAG  | 0.013966 | 0.005802 | 2.407300 | 0.205100 | 1.376000 | 0.204  | 0.002 | **  |
| V27NHHE | 0.012769 | 0.008435 | 1.513800 | 0.689200 | 1.646300 | 0.231  | 0.030 | *   |
| V3GF    | 0.012483 | 0.008488 | 1.470700 | 1.569300 | 0.646500 | 0.256  | 0.002 | **  |
| V4BTH   | 0.011472 | 0.009944 | 1.153700 | 1.946400 | 1.184200 | 0.280  | 0.116 |     |
| V36YFHE | 0.010708 | 0.008006 | 1.337500 | 1.121400 | 1.243700 | 0.302  | 0.100 | .   |
| V42SIL  | 0.010651 | 0.008608 | 1.237400 | 0.952800 | 0.420400 | 0.323  | 0.026 | *   |
| V13RWB  | 0.010585 | 0.007337 | 1.442700 | 1.080500 | 1.922100 | 0.345  | 0.159 |     |
| V33RWH  | 0.010302 | 0.006859 | 1.501900 | 1.318700 | 0.579800 | 0.366  | 0.025 | *   |
| V5GWH   | 0.010203 | 0.008334 | 1.224100 | 1.325200 | 0.674800 | 0.387  | 0.011 | *   |
| V16LR   | 0.009557 | 0.008123 | 1.176500 | 0.852900 | 1.184900 | 0.407  | 0.125 |     |
| V65LWB  | 0.009521 | 0.009391 | 1.013900 | 0.258700 | 0.741500 | 0.426  | 0.047 | *   |
| V21ESP  | 0.009484 | 0.007739 | 1.225400 | 1.236300 | 0.644000 | 0.445  | 0.147 |     |
| V23RFBT | 0.009142 | 0.006803 | 1.343900 | 0.990600 | 0.937700 | 0.464  | 0.264 |     |
| V63DWS  | 0.008333 | 0.007483 | 1.113600 | 0.313100 | 0.724800 | 0.481  | 0.019 | *   |
| V35STP  | 0.008318 | 0.007536 | 1.103700 | 0.700100 | 0.993200 | 0.498  | 0.350 |     |
| V7WEHE  | 0.008195 | 0.007490 | 1.094100 | 1.252500 | 1.241500 | 0.515  | 0.158 |     |
| V91NMIN | 0.008158 | 0.007918 | 1.030300 | 0.103300 | 0.719700 | 0.532  | 0.036 | *   |
| V19ER   | 0.007905 | 0.006671 | 1.184900 | 0.816000 | 1.308600 | 0.548  | 0.390 |     |

|          |          |          |          |          |          |       |       |   |
|----------|----------|----------|----------|----------|----------|-------|-------|---|
| V29CST   | 0.007715 | 0.005456 | 1.414100 | 0.548200 | 1.059100 | 0.564 | 0.077 | . |
| V56FTC   | 0.007610 | 0.006779 | 1.122500 | 0.948600 | 0.615700 | 0.579 | 0.114 |   |
| V14AUR   | 0.007440 | 0.007024 | 1.059200 | 0.893600 | 0.962100 | 0.594 | 0.164 |   |
| V81SHBC  | 0.007124 | 0.006361 | 1.120000 | 0.641600 | 0.271900 | 0.609 | 0.103 |   |
| V28VS    | 0.007070 | 0.008926 | 0.792000 | 0.599300 | 0.000000 | 0.624 | 0.911 |   |
| V12LK    | 0.006733 | 0.005709 | 1.179300 | 1.372500 | 0.801000 | 0.637 | 0.014 | * |
| V80RF    | 0.006689 | 0.007273 | 0.919800 | 0.568000 | 0.000000 | 0.651 | 0.238 |   |
| V50CHE   | 0.006645 | 0.008280 | 0.802400 | 0.580200 | 0.000000 | 0.664 | 0.313 |   |
| V46BHHE  | 0.006634 | 0.006911 | 0.959900 | 0.491200 | 0.307900 | 0.678 | 0.958 |   |
| V9SFW    | 0.006024 | 0.006604 | 0.912200 | 1.384400 | 1.734700 | 0.690 | 0.367 |   |
| V48YTBC  | 0.005842 | 0.007167 | 0.815200 | 0.511500 | 0.000000 | 0.702 | 0.386 |   |
| V41SPP   | 0.005728 | 0.005620 | 1.019200 | 1.058800 | 1.200500 | 0.714 | 0.721 |   |
| V8WNHE   | 0.005294 | 0.006131 | 0.863400 | 1.588600 | 1.843200 | 0.725 | 0.815 |   |
| V24BFCS  | 0.005165 | 0.005210 | 0.991300 | 0.751600 | 1.112500 | 0.736 | 0.569 |   |
| V37WHIP  | 0.005130 | 0.007087 | 0.723800 | 0.441900 | 0.000000 | 0.746 | 0.489 |   |
| V83SFC   | 0.005026 | 0.007077 | 0.710200 | 0.424600 | 0.000000 | 0.756 | 0.687 |   |
| V43GCU   | 0.004955 | 0.006908 | 0.717300 | 0.423100 | 0.000000 | 0.766 | 0.951 |   |
| V32SCC   | 0.004898 | 0.010846 | 0.451600 | 0.441300 | 0.000000 | 0.776 | 0.808 |   |
| V20PCU   | 0.004806 | 0.006607 | 0.727400 | 0.411800 | 0.000000 | 0.786 | 0.949 |   |
| V38GAL   | 0.004767 | 0.007107 | 0.670700 | 0.178700 | 0.323400 | 0.796 | 0.587 |   |
| V87LFC   | 0.004719 | 0.006574 | 0.717800 | 0.224900 | 0.289600 | 0.806 | 0.137 |   |
| V10WBSW  | 0.004538 | 0.005738 | 0.791000 | 1.426000 | 1.760400 | 0.815 | 0.996 |   |
| V2EYR    | 0.004523 | 0.005813 | 0.778100 | 1.285300 | 1.599500 | 0.824 | 0.859 |   |
| V53MB    | 0.004509 | 0.006393 | 0.705200 | 0.224200 | 0.250000 | 0.833 | 0.428 |   |
| V31AMAG  | 0.004040 | 0.007500 | 0.538600 | 0.370900 | 0.000000 | 0.842 | 0.778 |   |
| V55LHE   | 0.004024 | 0.006359 | 0.632900 | 0.354900 | 0.000000 | 0.850 | 0.468 |   |
| V45MGLK  | 0.003945 | 0.005906 | 0.668000 | 0.142600 | 0.271900 | 0.858 | 0.528 |   |
| V69SKF   | 0.003692 | 0.005819 | 0.634500 | 0.334500 | 0.000000 | 0.866 | 0.939 |   |
| V70RSL   | 0.003469 | 0.005750 | 0.603200 | 0.094600 | 0.236400 | 0.873 | 0.680 |   |
| V49LYRE  | 0.003229 | 0.005942 | 0.543500 | 0.268000 | 0.000000 | 0.879 | 0.646 |   |
| V85ROSE  | 0.002816 | 0.005168 | 0.545000 | 0.239700 | 0.000000 | 0.885 | 0.473 |   |
| V39FHE   | 0.002732 | 0.007692 | 0.355200 | 0.248500 | 0.000000 | 0.891 | 0.840 |   |
| V68PIL   | 0.002684 | 0.005897 | 0.455200 | 0.232900 | 0.000000 | 0.896 | 0.652 |   |
| V44MUSK  | 0.002563 | 0.007101 | 0.361000 | 0.236300 | 0.000000 | 0.901 | 0.734 |   |
| V98FLAME | 0.002557 | 0.005598 | 0.456700 | 0.220500 | 0.000000 | 0.907 | 0.712 |   |
| V51OWH   | 0.002408 | 0.005315 | 0.453100 | 0.211500 | 0.000000 | 0.911 | 0.410 |   |
| V16ST    | 0.002323 | 0.001704 | 1.363300 | 1.466600 | 1.493300 | 0.916 | 0.284 |   |
| V22SCR   | 0.002318 | 0.005111 | 0.453500 | 0.210400 | 0.000000 | 0.921 | 0.999 |   |
| V52TRM   | 0.002247 | 0.006427 | 0.349700 | 0.202800 | 0.000000 | 0.926 | 0.704 |   |
| V47RFC   | 0.002229 | 0.004899 | 0.455000 | 0.214300 | 0.000000 | 0.930 | 0.872 |   |
| V30BTR   | 0.002214 | 0.006274 | 0.352900 | 0.200700 | 0.000000 | 0.935 | 0.931 |   |
| V580B0   | 0.002124 | 0.004652 | 0.456500 | 0.198800 | 0.000000 | 0.939 | 0.983 |   |
| V26WWCH  | 0.002007 | 0.005550 | 0.361500 | 0.180300 | 0.000000 | 0.943 | 1.000 |   |
| V77RRP   | 0.001900 | 0.005290 | 0.359200 | 0.174100 | 0.000000 | 0.947 | 0.218 |   |
| V76GBB   | 0.001874 | 0.004111 | 0.455800 | 0.166700 | 0.000000 | 0.951 | 0.546 |   |
| V18YTH   | 0.001726 | 0.004872 | 0.354300 | 0.171800 | 0.000000 | 0.954 | 1.000 |   |
| V86BC00  | 0.001595 | 0.004421 | 0.360800 | 0.142300 | 0.000000 | 0.958 | 0.565 |   |
| V67GGC   | 0.001558 | 0.004376 | 0.356000 | 0.135400 | 0.000000 | 0.961 | 0.771 |   |
| V64BELL  | 0.001553 | 0.006272 | 0.247500 | 0.127500 | 0.000000 | 0.964 | 0.390 |   |
| V62RBTR  | 0.001522 | 0.004260 | 0.357300 | 0.127400 | 0.000000 | 0.967 | 0.609 |   |
| V57PINK  | 0.001480 | 0.004113 | 0.359800 | 0.127300 | 0.000000 | 0.970 | 0.362 |   |
| V92NFB   | 0.001412 | 0.003904 | 0.361800 | 0.141400 | 0.000000 | 0.973 | 0.583 |   |
| V97PCL   | 0.001237 | 0.004997 | 0.247500 | 0.102200 | 0.000000 | 0.976 | 0.108 |   |
| V40BRTH  | 0.001151 | 0.004652 | 0.247400 | 0.088800 | 0.000000 | 0.978 | 0.991 |   |
| V78LLOR  | 0.001038 | 0.004191 | 0.247800 | 0.106900 | 0.000000 | 0.980 | 0.129 |   |
| V60LFB   | 0.000929 | 0.003754 | 0.247500 | 0.076800 | 0.000000 | 0.982 | 0.925 |   |
| V84YRTH  | 0.000851 | 0.003434 | 0.247700 | 0.082700 | 0.000000 | 0.984 | 0.618 |   |
| V89PC00  | 0.000836 | 0.003378 | 0.247500 | 0.069100 | 0.000000 | 0.985 | 0.108 |   |
| V54STHR  | 0.000766 | 0.003093 | 0.247700 | 0.073200 | 0.000000 | 0.987 | 0.834 |   |
| V66CBW   | 0.000735 | 0.002968 | 0.247500 | 0.058800 | 0.000000 | 0.989 | 0.870 |   |
| V79YTHE  | 0.000733 | 0.002957 | 0.247800 | 0.075400 | 0.000000 | 0.990 | 0.636 |   |
| V96DF    | 0.000688 | 0.002777 | 0.247800 | 0.070800 | 0.000000 | 0.991 | 0.755 |   |
| V82AZKF  | 0.000680 | 0.002747 | 0.247500 | 0.053800 | 0.000000 | 0.993 | 0.111 |   |
| V100WBWS | 0.000643 | 0.002594 | 0.247800 | 0.066200 | 0.000000 | 0.994 | 0.129 |   |
| V93DB    | 0.000634 | 0.002557 | 0.247700 | 0.061600 | 0.000000 | 0.995 | 0.903 |   |

|          |          |          |          |          |          |       |       |
|----------|----------|----------|----------|----------|----------|-------|-------|
| V102KING | 0.000608 | 0.002454 | 0.247600 | 0.053800 | 0.000000 | 0.997 | 0.413 |
| V90WTG   | 0.000572 | 0.002311 | 0.247700 | 0.055600 | 0.000000 | 0.998 | 0.953 |
| V71PDOV  | 0.000540 | 0.002181 | 0.247800 | 0.055600 | 0.000000 | 0.999 | 0.767 |
| V94RBEE  | 0.000523 | 0.002110 | 0.247800 | 0.053800 | 0.000000 | 1.000 | 0.972 |
| V59YR    | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V61SPW   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V72CRP   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V73JW    | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V74BCHE  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V75RCR   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V88WG    | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V95HBC   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V99WWT   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V101LCOR | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |

---  
Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Contrast: Mixed\_Montane Fore

|         | average  | sd       | ratio    | ava      | avb      | cumsum | p        |
|---------|----------|----------|----------|----------|----------|--------|----------|
| V13RWB  | 0.009338 | 0.006981 | 1.337700 | 1.080500 | 0.662400 | 0.025  | 0.508    |
| V43GCU  | 0.009042 | 0.005499 | 1.644400 | 0.423100 | 1.256100 | 0.049  | 0.010 ** |
| V83SFC  | 0.008981 | 0.004949 | 1.814500 | 0.424600 | 1.220000 | 0.073  | 0.014 *  |
| V20PCU  | 0.008970 | 0.005919 | 1.515400 | 0.411800 | 1.268100 | 0.097  | 0.012 *  |
| V67GGC  | 0.008783 | 0.005683 | 1.545500 | 0.135400 | 0.926600 | 0.121  | 0.004 ** |
| V16LR   | 0.008555 | 0.006520 | 1.312100 | 0.852900 | 1.023200 | 0.144  | 0.324    |
| V46BHHE | 0.008352 | 0.007086 | 1.178600 | 0.491200 | 0.828800 | 0.166  | 0.540    |
| V33RWH  | 0.008286 | 0.006874 | 1.205500 | 1.318700 | 0.753700 | 0.188  | 0.215    |
| V37WHIP | 0.008217 | 0.006567 | 1.251200 | 0.441900 | 1.020400 | 0.211  | 0.031 *  |
| V23RBFT | 0.008056 | 0.006201 | 1.299200 | 0.990600 | 0.883700 | 0.232  | 0.736    |
| V36YFHE | 0.008026 | 0.007167 | 1.120000 | 1.121400 | 1.134500 | 0.254  | 0.873    |
| V28VS   | 0.007929 | 0.007261 | 1.091900 | 0.599300 | 0.709200 | 0.275  | 0.625    |
| V19ER   | 0.007909 | 0.006742 | 1.173100 | 0.816000 | 0.266900 | 0.296  | 0.368    |
| V48YTBC | 0.007759 | 0.006195 | 1.252500 | 0.511500 | 0.958400 | 0.317  | 0.046 *  |
| V54STHR | 0.007620 | 0.004661 | 1.634900 | 0.073200 | 0.758900 | 0.337  | 0.002 ** |
| V68PIL  | 0.007514 | 0.004974 | 1.510900 | 0.232900 | 0.785200 | 0.358  | 0.007 ** |
| V35STP  | 0.007322 | 0.006426 | 1.139400 | 0.700100 | 0.944200 | 0.377  | 0.776    |
| V27NHHE | 0.007298 | 0.008654 | 0.843300 | 0.689200 | 0.000000 | 0.397  | 0.686    |
| V49LYRE | 0.007170 | 0.005092 | 1.408100 | 0.268000 | 0.803100 | 0.416  | 0.008 ** |
| V50CHE  | 0.006930 | 0.005950 | 1.164700 | 0.580200 | 0.509600 | 0.435  | 0.286    |
| V80RF   | 0.006902 | 0.006031 | 1.144500 | 0.568000 | 0.913500 | 0.453  | 0.186    |
| V42SIL  | 0.006868 | 0.005994 | 1.145900 | 0.952800 | 1.363600 | 0.472  | 0.974    |
| V14AUR  | 0.006236 | 0.005868 | 1.062700 | 0.893600 | 0.919800 | 0.488  | 0.423    |
| V55LHE  | 0.006136 | 0.005947 | 1.031900 | 0.354900 | 0.590500 | 0.505  | 0.114    |
| V29CST  | 0.006120 | 0.006185 | 0.989500 | 0.548200 | 0.314400 | 0.521  | 0.884    |
| V24BFCS | 0.006092 | 0.005192 | 1.173200 | 0.751600 | 0.496700 | 0.537  | 0.254    |
| V26WWCH | 0.006052 | 0.009091 | 0.665800 | 0.180300 | 0.462400 | 0.554  | 0.965    |
| V17WPHE | 0.005965 | 0.009471 | 0.629800 | 0.602000 | 0.000000 | 0.570  | 0.986    |
| V81SHBC | 0.005961 | 0.005479 | 1.088100 | 0.641600 | 1.191300 | 0.586  | 0.856    |
| V8WNHE  | 0.005525 | 0.005093 | 1.084800 | 1.588600 | 1.709100 | 0.601  | 0.786    |
| V21ESP  | 0.005370 | 0.004726 | 1.136100 | 1.236300 | 1.502100 | 0.615  | 0.997    |
| V15STTH | 0.005290 | 0.006652 | 0.795300 | 1.475400 | 1.857500 | 0.629  | 0.948    |
| V86BC00 | 0.005217 | 0.005078 | 1.027400 | 0.142300 | 0.481700 | 0.643  | 0.053 .  |
| V11CR   | 0.004834 | 0.005601 | 0.863000 | 1.443400 | 1.710400 | 0.656  | 0.999    |
| V41SPP  | 0.004678 | 0.005784 | 0.808700 | 1.058800 | 1.352200 | 0.669  | 0.867    |
| V22SCR  | 0.004598 | 0.006404 | 0.718100 | 0.210400 | 0.317500 | 0.681  | 0.924    |
| V85ROSE | 0.004568 | 0.005803 | 0.787200 | 0.239700 | 0.331700 | 0.693  | 0.095 .  |
| V9SFW   | 0.004450 | 0.004930 | 0.902600 | 1.384400 | 1.525000 | 0.705  | 0.796    |
| V32SCC  | 0.004345 | 0.009627 | 0.451300 | 0.441300 | 0.000000 | 0.717  | 0.857    |
| V4BTH   | 0.004124 | 0.005531 | 0.745600 | 1.946400 | 2.294000 | 0.728  | 0.998    |
| V56FTC  | 0.004054 | 0.005016 | 0.808300 | 0.948600 | 1.231300 | 0.739  | 0.994    |
| V6WTTR  | 0.003848 | 0.005024 | 0.765900 | 1.325800 | 1.567100 | 0.749  | 0.973    |
| V62RBTR | 0.003800 | 0.005744 | 0.661600 | 0.127400 | 0.281200 | 0.759  | 0.141    |
| V7WEHE  | 0.003646 | 0.003640 | 1.001600 | 1.252500 | 1.224500 | 0.769  | 0.990    |
| V31AMAG | 0.003590 | 0.006655 | 0.539500 | 0.370900 | 0.000000 | 0.779  | 0.836    |

|          |          |          |          |          |          |       |       |
|----------|----------|----------|----------|----------|----------|-------|-------|
| V510WH   | 0.003548 | 0.004941 | 0.718000 | 0.211500 | 0.220000 | 0.788 | 0.237 |
| V56WH    | 0.003405 | 0.004259 | 0.799600 | 1.325200 | 1.486100 | 0.797 | 0.961 |
| V2EYR    | 0.003300 | 0.004546 | 0.725800 | 1.285300 | 1.440100 | 0.806 | 0.986 |
| V69SKF   | 0.003277 | 0.005158 | 0.635200 | 0.334500 | 0.000000 | 0.815 | 0.969 |
| V57PINK  | 0.003208 | 0.004800 | 0.668200 | 0.127300 | 0.250000 | 0.824 | 0.145 |
| V76GBB   | 0.003203 | 0.004482 | 0.714700 | 0.166700 | 0.220000 | 0.832 | 0.173 |
| V63DWS   | 0.003119 | 0.004987 | 0.625500 | 0.313100 | 0.000000 | 0.840 | 0.836 |
| V10WBSW  | 0.002993 | 0.004868 | 0.614800 | 1.426000 | 1.622600 | 0.849 | 1.000 |
| V36F     | 0.002942 | 0.004175 | 0.704800 | 1.569300 | 1.724600 | 0.856 | 0.987 |
| V65LWB   | 0.002877 | 0.008006 | 0.359400 | 0.258700 | 0.000000 | 0.864 | 0.613 |
| V102KING | 0.002549 | 0.004128 | 0.617500 | 0.053800 | 0.236400 | 0.871 | 0.113 |
| V39FHE   | 0.002424 | 0.006810 | 0.356000 | 0.248500 | 0.000000 | 0.877 | 0.862 |
| V34WSW   | 0.002385 | 0.005233 | 0.455700 | 0.247500 | 0.000000 | 0.884 | 0.994 |
| V53MB    | 0.002366 | 0.005345 | 0.442500 | 0.224200 | 0.000000 | 0.890 | 0.904 |
| V44MUSK  | 0.002278 | 0.006306 | 0.361200 | 0.236300 | 0.000000 | 0.896 | 0.779 |
| V98FLAME | 0.002256 | 0.004937 | 0.457000 | 0.220500 | 0.000000 | 0.902 | 0.806 |
| V87LFC   | 0.002212 | 0.004906 | 0.450900 | 0.224900 | 0.000000 | 0.908 | 0.687 |
| V52TRM   | 0.001993 | 0.005682 | 0.350700 | 0.202800 | 0.000000 | 0.914 | 0.792 |
| V47RFC   | 0.001992 | 0.004375 | 0.455300 | 0.214300 | 0.000000 | 0.919 | 0.926 |
| V30BTR   | 0.001964 | 0.005551 | 0.353800 | 0.200700 | 0.000000 | 0.924 | 0.931 |
| V25WAG   | 0.001953 | 0.004286 | 0.455700 | 0.205100 | 0.000000 | 0.929 | 0.997 |
| V580B0   | 0.001891 | 0.004139 | 0.456800 | 0.198800 | 0.000000 | 0.934 | 0.990 |
| V16ST    | 0.001754 | 0.001220 | 1.437400 | 1.466600 | 1.379200 | 0.939 | 0.861 |
| V38GAL   | 0.001728 | 0.004792 | 0.360600 | 0.178700 | 0.000000 | 0.944 | 0.969 |
| V77RRP   | 0.001687 | 0.004691 | 0.359700 | 0.174100 | 0.000000 | 0.948 | 0.386 |
| V18YTH   | 0.001548 | 0.004368 | 0.354500 | 0.171800 | 0.000000 | 0.953 | 0.999 |
| V45MGLK  | 0.001409 | 0.003896 | 0.361600 | 0.142600 | 0.000000 | 0.956 | 0.975 |
| V64BELL  | 0.001363 | 0.005506 | 0.247500 | 0.127500 | 0.000000 | 0.960 | 0.558 |
| V92NFB   | 0.001268 | 0.003504 | 0.361700 | 0.141400 | 0.000000 | 0.963 | 0.790 |
| V12LK    | 0.001220 | 0.000891 | 1.368900 | 1.372500 | 1.324100 | 0.967 | 0.997 |
| V97PCL   | 0.001087 | 0.004390 | 0.247500 | 0.102200 | 0.000000 | 0.970 | 0.485 |
| V70RSL   | 0.001006 | 0.004066 | 0.247500 | 0.094600 | 0.000000 | 0.972 | 0.990 |
| V40BRTH  | 0.001003 | 0.004051 | 0.247500 | 0.088800 | 0.000000 | 0.975 | 0.997 |
| V91NMIN  | 0.000978 | 0.002735 | 0.357500 | 0.103300 | 0.000000 | 0.978 | 0.971 |
| V78LLOR  | 0.000935 | 0.003773 | 0.247700 | 0.106900 | 0.000000 | 0.980 | 0.519 |
| V60LFB   | 0.000816 | 0.003298 | 0.247500 | 0.076800 | 0.000000 | 0.982 | 0.960 |
| V84YRTH  | 0.000761 | 0.003073 | 0.247700 | 0.082700 | 0.000000 | 0.984 | 0.792 |
| V89PC00  | 0.000734 | 0.002967 | 0.247500 | 0.069100 | 0.000000 | 0.986 | 0.485 |
| V79YTHE  | 0.000659 | 0.002662 | 0.247700 | 0.075400 | 0.000000 | 0.988 | 0.809 |
| V66CBW   | 0.000643 | 0.002597 | 0.247500 | 0.058800 | 0.000000 | 0.990 | 0.918 |
| V96DF    | 0.000619 | 0.002499 | 0.247700 | 0.070800 | 0.000000 | 0.991 | 0.823 |
| V82AZKF  | 0.000594 | 0.002400 | 0.247500 | 0.053800 | 0.000000 | 0.993 | 0.505 |
| V100WBWS | 0.000579 | 0.002335 | 0.247700 | 0.066200 | 0.000000 | 0.995 | 0.519 |
| V93DB    | 0.000567 | 0.002289 | 0.247700 | 0.061600 | 0.000000 | 0.996 | 0.970 |
| V90WTG   | 0.000512 | 0.002068 | 0.247700 | 0.055600 | 0.000000 | 0.997 | 0.978 |
| V71PDOV  | 0.000486 | 0.001964 | 0.247700 | 0.055600 | 0.000000 | 0.999 | 0.909 |
| V94RBEE  | 0.000470 | 0.001899 | 0.247700 | 0.053800 | 0.000000 | 1.000 | 0.985 |
| V59YR    | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V61SPW   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V72CRP   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V73JW    | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V74BCHE  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V75RCR   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V88WG    | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V95HBC   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V99WWT   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V101LCOR | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |

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Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Contrast: Mixed\_Foothills Wo

|         | average  | sd       | ratio    | ava      | avb      | cumsum | p         |
|---------|----------|----------|----------|----------|----------|--------|-----------|
| V40BRTH | 0.014392 | 0.009230 | 1.559200 | 0.088800 | 1.423800 | 0.037  | 0.006 **  |
| V22SCR  | 0.012437 | 0.004863 | 2.557800 | 0.210400 | 1.364500 | 0.068  | 0.001 *** |



|          |          |          |          |          |          |       |       |   |
|----------|----------|----------|----------|----------|----------|-------|-------|---|
| V26WWCH  | 0.011907 | 0.007887 | 1.509700 | 0.180300 | 1.210300 | 0.099 | 0.013 | * |
| V18YTH   | 0.011886 | 0.007877 | 1.508900 | 0.171800 | 1.199300 | 0.129 | 0.036 | * |
| V10WBSW  | 0.009520 | 0.007570 | 1.257700 | 1.426000 | 0.773600 | 0.154 | 0.253 |   |
| V23RBFT  | 0.009458 | 0.007442 | 1.270900 | 0.990600 | 0.243500 | 0.178 | 0.178 |   |
| V28VS    | 0.009159 | 0.007107 | 1.288800 | 0.599300 | 1.088900 | 0.201 | 0.144 |   |
| V53MB    | 0.008400 | 0.005483 | 1.532100 | 0.224200 | 0.851600 | 0.222 | 0.015 | * |
| V46BHHE  | 0.008159 | 0.006034 | 1.352200 | 0.491200 | 0.931700 | 0.243 | 0.631 |   |
| V16LR    | 0.008067 | 0.006758 | 1.193700 | 0.852900 | 1.479800 | 0.264 | 0.512 |   |
| V19ER    | 0.007778 | 0.006808 | 1.142400 | 0.816000 | 0.654600 | 0.284 | 0.431 |   |
| V69SKF   | 0.007708 | 0.006107 | 1.262200 | 0.334500 | 0.890600 | 0.304 | 0.054 | . |
| V35STP   | 0.007702 | 0.006775 | 1.136900 | 0.700100 | 1.011200 | 0.323 | 0.634 |   |
| V42SIL   | 0.007673 | 0.005328 | 1.440100 | 0.952800 | 0.761200 | 0.343 | 0.805 |   |
| V27NHHE  | 0.007657 | 0.007842 | 0.976500 | 0.689200 | 0.256000 | 0.362 | 0.631 |   |
| V50CHE   | 0.007225 | 0.007685 | 0.940200 | 0.580200 | 0.384600 | 0.381 | 0.206 |   |
| V36YFHE  | 0.007150 | 0.005676 | 1.259600 | 1.121400 | 1.319700 | 0.399 | 0.979 |   |
| V29CST   | 0.007090 | 0.006413 | 1.105700 | 0.548200 | 0.659900 | 0.417 | 0.281 |   |
| V20PCU   | 0.007080 | 0.006583 | 1.075500 | 0.411800 | 0.673000 | 0.435 | 0.211 |   |
| V9SFW    | 0.007048 | 0.006634 | 1.062400 | 1.384400 | 1.081900 | 0.453 | 0.222 |   |
| V14AUR   | 0.006603 | 0.006170 | 1.070200 | 0.893600 | 0.987000 | 0.470 | 0.313 |   |
| V43GCU   | 0.006573 | 0.006112 | 1.075300 | 0.423100 | 0.583000 | 0.487 | 0.467 |   |
| V80RF    | 0.006348 | 0.006410 | 0.990200 | 0.568000 | 0.304500 | 0.503 | 0.351 |   |
| V13RWB   | 0.006090 | 0.004647 | 1.310500 | 1.080500 | 1.170600 | 0.519 | 0.999 |   |
| V17WPHE  | 0.006088 | 0.009642 | 0.631400 | 0.602000 | 0.000000 | 0.534 | 0.987 |   |
| V64BELL  | 0.005858 | 0.009467 | 0.618800 | 0.127500 | 0.492000 | 0.549 | 0.144 |   |
| V81SHBC  | 0.005798 | 0.005361 | 1.081400 | 0.641600 | 0.807100 | 0.564 | 0.929 |   |
| V15STTH  | 0.005476 | 0.006465 | 0.847000 | 1.475400 | 1.788200 | 0.578 | 0.957 |   |
| V83SFC   | 0.005382 | 0.006098 | 0.882700 | 0.424600 | 0.261700 | 0.592 | 0.563 |   |
| V24BFCS  | 0.005280 | 0.004956 | 1.065500 | 0.751600 | 0.794400 | 0.605 | 0.471 |   |
| V41SPP   | 0.005277 | 0.006727 | 0.784500 | 1.058800 | 1.526000 | 0.619 | 0.796 |   |
| V48YBTC  | 0.005274 | 0.006448 | 0.817800 | 0.511500 | 0.000000 | 0.632 | 0.488 |   |
| V8WNHE   | 0.005251 | 0.004942 | 1.062400 | 1.588600 | 1.582200 | 0.645 | 0.812 |   |
| V11CR    | 0.004856 | 0.004784 | 1.015100 | 1.443400 | 1.484400 | 0.658 | 0.999 |   |
| V21ESP   | 0.004753 | 0.004815 | 0.987100 | 1.236300 | 1.346100 | 0.670 | 1.000 |   |
| V37WHIP  | 0.004623 | 0.006367 | 0.726100 | 0.441900 | 0.000000 | 0.682 | 0.661 |   |
| V32SCC   | 0.004434 | 0.009801 | 0.452400 | 0.441300 | 0.000000 | 0.693 | 0.852 |   |
| V63DWS   | 0.004331 | 0.005235 | 0.827300 | 0.313100 | 0.243500 | 0.704 | 0.596 |   |
| V7WEHE   | 0.004274 | 0.004093 | 1.044200 | 1.252500 | 1.318200 | 0.715 | 0.960 |   |
| V56FTC   | 0.004240 | 0.005159 | 0.821900 | 0.948600 | 1.244700 | 0.726 | 0.990 |   |
| V33RWH   | 0.004130 | 0.005045 | 0.818600 | 1.318700 | 1.597500 | 0.736 | 0.991 |   |
| V4BTH    | 0.004096 | 0.004957 | 0.826300 | 1.946400 | 1.975000 | 0.747 | 0.997 |   |
| V96DF    | 0.003970 | 0.006435 | 0.617000 | 0.070800 | 0.349000 | 0.757 | 0.083 | . |
| V580B0   | 0.003798 | 0.005307 | 0.715600 | 0.198800 | 0.261700 | 0.767 | 0.832 |   |
| V6WTR    | 0.003785 | 0.005196 | 0.728400 | 1.325800 | 1.579900 | 0.776 | 0.983 |   |
| V2EYR    | 0.003733 | 0.004281 | 0.872000 | 1.285300 | 1.316800 | 0.786 | 0.971 |   |
| V31AMAG  | 0.003663 | 0.006775 | 0.540700 | 0.370900 | 0.000000 | 0.795 | 0.824 |   |
| V55LHE   | 0.003636 | 0.005729 | 0.634700 | 0.354900 | 0.000000 | 0.804 | 0.551 |   |
| V56WH    | 0.003634 | 0.003729 | 0.974400 | 1.325200 | 1.309900 | 0.814 | 0.930 |   |
| V61SPW   | 0.003414 | 0.005979 | 0.571000 | 0.000000 | 0.326200 | 0.823 | 0.219 |   |
| V36F     | 0.003253 | 0.003748 | 0.868100 | 1.569300 | 1.569600 | 0.831 | 0.979 |   |
| V62RBTR  | 0.003142 | 0.004683 | 0.670900 | 0.127400 | 0.220000 | 0.839 | 0.289 |   |
| V65LWB   | 0.002942 | 0.008165 | 0.360300 | 0.258700 | 0.000000 | 0.846 | 0.611 |   |
| V49LYRE  | 0.002899 | 0.005317 | 0.545200 | 0.268000 | 0.000000 | 0.854 | 0.701 |   |
| V66CBW   | 0.002697 | 0.004351 | 0.619700 | 0.058800 | 0.220000 | 0.861 | 0.395 |   |
| V85ROSE  | 0.002535 | 0.004640 | 0.546400 | 0.239700 | 0.000000 | 0.867 | 0.502 |   |
| V39FHE   | 0.002474 | 0.006936 | 0.356600 | 0.248500 | 0.000000 | 0.874 | 0.883 |   |
| V34WSW   | 0.002433 | 0.005328 | 0.456700 | 0.247500 | 0.000000 | 0.880 | 0.993 |   |
| V16ST    | 0.002432 | 0.002103 | 1.156600 | 1.466600 | 1.577800 | 0.886 | 0.244 |   |
| V68PIL   | 0.002421 | 0.005305 | 0.456500 | 0.232900 | 0.000000 | 0.892 | 0.689 |   |
| V44MUSK  | 0.002324 | 0.006421 | 0.361900 | 0.236300 | 0.000000 | 0.898 | 0.800 |   |
| V98FLAME | 0.002304 | 0.005029 | 0.458100 | 0.220500 | 0.000000 | 0.904 | 0.801 |   |
| V87LFC   | 0.002258 | 0.004995 | 0.452000 | 0.224900 | 0.000000 | 0.910 | 0.668 |   |
| V51OWH   | 0.002175 | 0.004792 | 0.453900 | 0.211500 | 0.000000 | 0.915 | 0.413 |   |
| V52TRM   | 0.002033 | 0.005787 | 0.351300 | 0.202800 | 0.000000 | 0.920 | 0.789 |   |
| V47RFC   | 0.002030 | 0.004451 | 0.456200 | 0.214300 | 0.000000 | 0.926 | 0.916 |   |
| V30BTR   | 0.002004 | 0.005654 | 0.354500 | 0.200700 | 0.000000 | 0.931 | 0.941 |   |

|          |          |          |          |          |          |       |       |
|----------|----------|----------|----------|----------|----------|-------|-------|
| V25WAG   | 0.001992 | 0.004362 | 0.456700 | 0.205100 | 0.000000 | 0.936 | 0.997 |
| V38GAL   | 0.001763 | 0.004879 | 0.361300 | 0.178700 | 0.000000 | 0.940 | 0.969 |
| V77RRP   | 0.001722 | 0.004777 | 0.360400 | 0.174100 | 0.000000 | 0.945 | 0.385 |
| V76GBB   | 0.001693 | 0.003703 | 0.457300 | 0.166700 | 0.000000 | 0.949 | 0.674 |
| V86BC00  | 0.001443 | 0.003991 | 0.361600 | 0.142300 | 0.000000 | 0.953 | 0.610 |
| V45MGLK  | 0.001438 | 0.003967 | 0.362300 | 0.142600 | 0.000000 | 0.956 | 0.970 |
| V12LK    | 0.001432 | 0.001053 | 1.360800 | 1.372500 | 1.350100 | 0.960 | 0.979 |
| V67GGC   | 0.001406 | 0.003942 | 0.356500 | 0.135400 | 0.000000 | 0.964 | 0.870 |
| V57PINK  | 0.001334 | 0.003700 | 0.360500 | 0.127300 | 0.000000 | 0.967 | 0.455 |
| V92NFB   | 0.001291 | 0.003564 | 0.362400 | 0.141400 | 0.000000 | 0.970 | 0.762 |
| V97PCL   | 0.001110 | 0.004475 | 0.248100 | 0.102200 | 0.000000 | 0.973 | 0.429 |
| V70RSL   | 0.001028 | 0.004145 | 0.248100 | 0.094600 | 0.000000 | 0.976 | 0.989 |
| V91NMIN  | 0.000997 | 0.002784 | 0.358300 | 0.103300 | 0.000000 | 0.978 | 0.976 |
| V78LLOR  | 0.000952 | 0.003836 | 0.248100 | 0.106900 | 0.000000 | 0.981 | 0.456 |
| V60LFB   | 0.000834 | 0.003362 | 0.248100 | 0.076800 | 0.000000 | 0.983 | 0.961 |
| V84YRTH  | 0.000776 | 0.003127 | 0.248100 | 0.082700 | 0.000000 | 0.985 | 0.763 |
| V89PC00  | 0.000750 | 0.003025 | 0.248100 | 0.069100 | 0.000000 | 0.987 | 0.428 |
| V54STHR  | 0.000698 | 0.002812 | 0.248100 | 0.073200 | 0.000000 | 0.989 | 0.855 |
| V79YTHE  | 0.000672 | 0.002706 | 0.248100 | 0.075400 | 0.000000 | 0.990 | 0.763 |
| V82AZKF  | 0.000607 | 0.002448 | 0.248100 | 0.053800 | 0.000000 | 0.992 | 0.414 |
| V100WBWS | 0.000589 | 0.002375 | 0.248100 | 0.066200 | 0.000000 | 0.993 | 0.456 |
| V93DB    | 0.000578 | 0.002329 | 0.248100 | 0.061600 | 0.000000 | 0.995 | 0.961 |
| V102KING | 0.000549 | 0.002214 | 0.248100 | 0.053800 | 0.000000 | 0.996 | 0.537 |
| V90WTG   | 0.000522 | 0.002104 | 0.248100 | 0.055600 | 0.000000 | 0.998 | 0.980 |
| V71PD0V  | 0.000495 | 0.001997 | 0.248100 | 0.055600 | 0.000000 | 0.999 | 0.898 |
| V94RBEE  | 0.000479 | 0.001931 | 0.248100 | 0.053800 | 0.000000 | 1.000 | 0.989 |
| V59YR    | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V72CRP   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V73JW    | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V74BCHE  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V75RCR   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V88WG    | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V95HBC   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V99WWT   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V101LCOR | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |

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Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Contrast: Mixed\_Box-Ironbark

|         | average  | sd       | ratio    | ava      | avb      | cumsum | p         |
|---------|----------|----------|----------|----------|----------|--------|-----------|
| V18YTH  | 0.018658 | 0.005852 | 3.188000 | 0.171800 | 1.909900 | 0.039  | 0.001 *** |
| V26WWCH | 0.015902 | 0.005536 | 2.872000 | 0.180300 | 1.655500 | 0.072  | 0.001 *** |
| V10WBSW | 0.015288 | 0.006045 | 2.529000 | 1.426000 | 0.000000 | 0.103  | 0.002 **  |
| V39FHE  | 0.014967 | 0.004566 | 3.278000 | 0.248500 | 1.532700 | 0.134  | 0.001 *** |
| V46BHHE | 0.014098 | 0.007088 | 1.989000 | 0.491200 | 1.798900 | 0.163  | 0.002 **  |
| V40BRTH | 0.013699 | 0.008353 | 1.640000 | 0.088800 | 1.379300 | 0.192  | 0.010 **  |
| V21ESP  | 0.013293 | 0.006583 | 2.019000 | 1.236300 | 0.000000 | 0.219  | 0.001 *** |
| V22SCR  | 0.012484 | 0.005255 | 2.376000 | 0.210400 | 1.350800 | 0.245  | 0.002 **  |
| V44MUSK | 0.011843 | 0.007846 | 1.509000 | 0.236300 | 1.180800 | 0.269  | 0.001 *** |
| V75RCR  | 0.010748 | 0.003075 | 3.495000 | 0.000000 | 1.000900 | 0.292  | 0.001 *** |
| V580B0  | 0.009787 | 0.004514 | 2.168000 | 0.198800 | 1.059200 | 0.312  | 0.002 **  |
| V23RBFT | 0.009490 | 0.007770 | 1.221000 | 0.990600 | 0.320500 | 0.332  | 0.164     |
| V13RWB  | 0.008878 | 0.006402 | 1.387000 | 1.080500 | 1.792200 | 0.350  | 0.669     |
| V56FTC  | 0.008670 | 0.006269 | 1.383000 | 0.948600 | 0.281200 | 0.368  | 0.028 *   |
| V28VS   | 0.008644 | 0.006295 | 1.373000 | 0.599300 | 0.971800 | 0.386  | 0.304     |
| V11CR   | 0.008437 | 0.007081 | 1.192000 | 1.443400 | 0.992900 | 0.403  | 0.757     |
| V36YFHE | 0.008272 | 0.006141 | 1.347000 | 1.121400 | 1.383500 | 0.420  | 0.827     |
| V16LR   | 0.008079 | 0.006977 | 1.158000 | 0.852900 | 0.576500 | 0.437  | 0.530     |
| V35STP  | 0.007717 | 0.006654 | 1.160000 | 0.700100 | 0.656700 | 0.453  | 0.662     |
| V27NHHE | 0.007497 | 0.008904 | 0.842000 | 0.689200 | 0.000000 | 0.468  | 0.640     |
| V19ER   | 0.007343 | 0.005945 | 1.235000 | 0.816000 | 0.846400 | 0.484  | 0.660     |
| V43GCU  | 0.007068 | 0.005570 | 1.269000 | 0.423100 | 0.818400 | 0.498  | 0.279     |
| V42SIL  | 0.006994 | 0.005548 | 1.261000 | 0.952800 | 1.255000 | 0.513  | 0.945     |
| V4BTH   | 0.006788 | 0.003994 | 1.699000 | 1.946400 | 1.520900 | 0.527  | 0.833     |

|          |          |          |          |          |          |       |         |
|----------|----------|----------|----------|----------|----------|-------|---------|
| V2EYR    | 0.006709 | 0.005999 | 1.118000 | 1.285300 | 0.874900 | 0.541 | 0.412   |
| V81SHBC  | 0.006489 | 0.005846 | 1.110000 | 0.641600 | 0.281200 | 0.554 | 0.426   |
| V66CBW   | 0.006361 | 0.006379 | 0.997000 | 0.058800 | 0.566600 | 0.567 | 0.012 * |
| V70RSL   | 0.006278 | 0.006199 | 1.013000 | 0.094600 | 0.557400 | 0.580 | 0.164   |
| V17WPHE  | 0.006117 | 0.009722 | 0.629000 | 0.602000 | 0.000000 | 0.593 | 0.989   |
| V80RF    | 0.006050 | 0.006578 | 0.920000 | 0.568000 | 0.000000 | 0.605 | 0.480   |
| V90WTG   | 0.006046 | 0.006464 | 0.935000 | 0.055600 | 0.542000 | 0.618 | 0.041 * |
| V50CHE   | 0.006025 | 0.007514 | 0.802000 | 0.580200 | 0.000000 | 0.630 | 0.480   |
| V29CST   | 0.005983 | 0.006099 | 0.981000 | 0.548200 | 0.256000 | 0.643 | 0.931   |
| V61SPW   | 0.005859 | 0.005982 | 0.980000 | 0.000000 | 0.593000 | 0.655 | 0.025 * |
| V14AUR   | 0.005388 | 0.005537 | 0.973000 | 0.893600 | 1.211600 | 0.666 | 0.682   |
| V48YTBC  | 0.005300 | 0.006506 | 0.815000 | 0.511500 | 0.000000 | 0.677 | 0.559   |
| V15STTH  | 0.005211 | 0.006022 | 0.865000 | 1.475400 | 1.627000 | 0.688 | 0.966   |
| V9SFW    | 0.005137 | 0.005014 | 1.024000 | 1.384400 | 1.472800 | 0.698 | 0.620   |
| V41SPP   | 0.004977 | 0.006330 | 0.786000 | 1.058800 | 1.440100 | 0.709 | 0.841   |
| V30BTR   | 0.004843 | 0.007338 | 0.660000 | 0.200700 | 0.382900 | 0.719 | 0.573   |
| V56WH    | 0.004733 | 0.003669 | 1.290000 | 1.325200 | 1.171900 | 0.729 | 0.717   |
| V36F     | 0.004725 | 0.003539 | 1.335000 | 1.569300 | 1.330200 | 0.738 | 0.722   |
| V8WNHE   | 0.004707 | 0.005030 | 0.936000 | 1.588600 | 1.595400 | 0.748 | 0.911   |
| V37WHIP  | 0.004647 | 0.006424 | 0.723000 | 0.441900 | 0.000000 | 0.758 | 0.671   |
| V83SFC   | 0.004543 | 0.006404 | 0.710000 | 0.424600 | 0.000000 | 0.767 | 0.785   |
| V24BFCS  | 0.004457 | 0.004309 | 1.034000 | 0.751600 | 1.014400 | 0.776 | 0.806   |
| V32SCC   | 0.004454 | 0.009873 | 0.451000 | 0.441300 | 0.000000 | 0.785 | 0.831   |
| V20PCU   | 0.004351 | 0.005983 | 0.727000 | 0.411800 | 0.000000 | 0.794 | 0.977   |
| V79YTHE  | 0.004127 | 0.006650 | 0.621000 | 0.075400 | 0.396100 | 0.803 | 0.032 * |
| V63DWS   | 0.004112 | 0.005037 | 0.816000 | 0.313100 | 0.220000 | 0.811 | 0.629   |
| V7WEHE   | 0.004030 | 0.003983 | 1.012000 | 1.252500 | 1.296700 | 0.820 | 0.978   |
| V33RWH   | 0.003758 | 0.004830 | 0.778000 | 1.318700 | 1.512200 | 0.828 | 0.998   |
| V6WTR    | 0.003705 | 0.004996 | 0.741000 | 1.325800 | 1.525100 | 0.835 | 0.984   |
| V31AMAG  | 0.003680 | 0.006826 | 0.539000 | 0.370900 | 0.000000 | 0.843 | 0.803   |
| V55LHE   | 0.003654 | 0.005776 | 0.633000 | 0.354900 | 0.000000 | 0.850 | 0.591   |
| V69SKF   | 0.003359 | 0.005292 | 0.635000 | 0.334500 | 0.000000 | 0.857 | 0.964   |
| V76GBB   | 0.003230 | 0.004486 | 0.720000 | 0.166700 | 0.210200 | 0.864 | 0.170   |
| V65LWB   | 0.002960 | 0.008240 | 0.359000 | 0.258700 | 0.000000 | 0.870 | 0.592   |
| V49LYRE  | 0.002915 | 0.005365 | 0.543000 | 0.268000 | 0.000000 | 0.876 | 0.686   |
| V96DF    | 0.002818 | 0.004531 | 0.622000 | 0.070800 | 0.256000 | 0.882 | 0.228   |
| V12LK    | 0.002704 | 0.001574 | 1.718000 | 1.372500 | 1.122800 | 0.888 | 0.610   |
| V94RBEE  | 0.002692 | 0.004346 | 0.620000 | 0.053800 | 0.256000 | 0.893 | 0.405   |
| V85ROSE  | 0.002548 | 0.004679 | 0.545000 | 0.239700 | 0.000000 | 0.898 | 0.546   |
| V34WSW   | 0.002444 | 0.005368 | 0.455000 | 0.247500 | 0.000000 | 0.903 | 0.997   |
| V68PTL   | 0.002434 | 0.005347 | 0.455000 | 0.232900 | 0.000000 | 0.908 | 0.689   |
| V53MB    | 0.002430 | 0.005496 | 0.442000 | 0.224200 | 0.000000 | 0.914 | 0.902   |
| V16ST    | 0.002363 | 0.001590 | 1.487000 | 1.466600 | 1.425700 | 0.918 | 0.293   |
| V98FLAME | 0.002315 | 0.005070 | 0.457000 | 0.220500 | 0.000000 | 0.923 | 0.789   |
| V87LFC   | 0.002268 | 0.005033 | 0.451000 | 0.224900 | 0.000000 | 0.928 | 0.623   |
| V510WH   | 0.002186 | 0.004829 | 0.453000 | 0.211500 | 0.000000 | 0.932 | 0.457   |
| V88WG    | 0.002185 | 0.003827 | 0.571000 | 0.000000 | 0.210200 | 0.937 | 0.277   |
| V95HBC   | 0.002185 | 0.003827 | 0.571000 | 0.000000 | 0.210200 | 0.942 | 0.136   |
| V52TRM   | 0.002043 | 0.005832 | 0.350000 | 0.202800 | 0.000000 | 0.946 | 0.772   |
| V47RFC   | 0.002039 | 0.004481 | 0.455000 | 0.214300 | 0.000000 | 0.950 | 0.905   |
| V25WAG   | 0.002001 | 0.004394 | 0.455000 | 0.205100 | 0.000000 | 0.954 | 0.999   |
| V38GAL   | 0.001771 | 0.004914 | 0.360000 | 0.178700 | 0.000000 | 0.958 | 0.979   |
| V77RRP   | 0.001730 | 0.004812 | 0.359000 | 0.174100 | 0.000000 | 0.961 | 0.342   |
| V86BC00  | 0.001450 | 0.004021 | 0.361000 | 0.142300 | 0.000000 | 0.964 | 0.588   |
| V45MGLK  | 0.001444 | 0.003997 | 0.361000 | 0.142600 | 0.000000 | 0.967 | 0.977   |
| V67GGC   | 0.001413 | 0.003972 | 0.356000 | 0.135400 | 0.000000 | 0.970 | 0.841   |
| V64BELL  | 0.001400 | 0.005660 | 0.247000 | 0.127500 | 0.000000 | 0.973 | 0.536   |
| V62RBTR  | 0.001374 | 0.003843 | 0.358000 | 0.127400 | 0.000000 | 0.976 | 0.661   |
| V57PINK  | 0.001341 | 0.003729 | 0.359000 | 0.127300 | 0.000000 | 0.979 | 0.479   |
| V92NFB   | 0.001296 | 0.003585 | 0.362000 | 0.141400 | 0.000000 | 0.981 | 0.743   |
| V97PCL   | 0.001116 | 0.004512 | 0.247000 | 0.102200 | 0.000000 | 0.984 | 0.402   |
| V91NMN   | 0.001002 | 0.002802 | 0.357000 | 0.103300 | 0.000000 | 0.986 | 0.973   |
| V78LLOR  | 0.000955 | 0.003858 | 0.248000 | 0.106900 | 0.000000 | 0.988 | 0.412   |
| V60LFB   | 0.000839 | 0.003390 | 0.247000 | 0.076800 | 0.000000 | 0.990 | 0.963   |
| V84YRTH  | 0.000779 | 0.003146 | 0.248000 | 0.082700 | 0.000000 | 0.991 | 0.755   |

|          |          |          |          |          |          |       |       |
|----------|----------|----------|----------|----------|----------|-------|-------|
| V89PC00  | 0.000755 | 0.003050 | 0.247000 | 0.069100 | 0.000000 | 0.993 | 0.402 |
| V54STHR  | 0.000701 | 0.002830 | 0.248000 | 0.073200 | 0.000000 | 0.994 | 0.864 |
| V82AZKF  | 0.000611 | 0.002470 | 0.247000 | 0.053800 | 0.000000 | 0.995 | 0.410 |
| V100WBWS | 0.000591 | 0.002388 | 0.248000 | 0.066200 | 0.000000 | 0.997 | 0.412 |
| V93DB    | 0.000580 | 0.002343 | 0.248000 | 0.061600 | 0.000000 | 0.998 | 0.960 |
| V102KING | 0.000552 | 0.002230 | 0.247000 | 0.053800 | 0.000000 | 0.999 | 0.569 |
| V71PDOV  | 0.000497 | 0.002008 | 0.248000 | 0.055600 | 0.000000 | 1.000 | 0.881 |
| V59YR    | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V72CRP   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V73JW    | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V74BCHE  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V99WWT   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V101LCOR | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |

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Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Contrast: Mixed\_River Red Gu

|          | average  | sd       | ratio    | ava      | avb      | cumsum | p     |     |
|----------|----------|----------|----------|----------|----------|--------|-------|-----|
| V4BTH    | 0.020445 | 0.006218 | 3.288000 | 1.946400 | 0.000000 | 0.030  | 0.001 | *** |
| V32SCC   | 0.020280 | 0.007521 | 2.697000 | 0.441300 | 2.225000 | 0.060  | 0.001 | *** |
| V17WPHE  | 0.015917 | 0.009167 | 1.736000 | 0.602000 | 2.034900 | 0.084  | 0.018 | *   |
| V10WBSW  | 0.015087 | 0.005860 | 2.574000 | 1.426000 | 0.000000 | 0.106  | 0.001 | *** |
| V11CR    | 0.015054 | 0.006597 | 2.282000 | 1.443400 | 0.000000 | 0.128  | 0.001 | *** |
| V38GAL   | 0.014937 | 0.005417 | 2.758000 | 0.178700 | 1.582800 | 0.151  | 0.001 | *** |
| V26WWCH  | 0.014881 | 0.004857 | 3.064000 | 0.180300 | 1.578000 | 0.173  | 0.002 | **  |
| V8WNHE   | 0.014481 | 0.007804 | 1.856000 | 1.588600 | 0.289600 | 0.194  | 0.004 | **  |
| V30BTR   | 0.014462 | 0.004730 | 3.057000 | 0.200700 | 1.520300 | 0.216  | 0.001 | *** |
| V2EYR    | 0.013450 | 0.005370 | 2.505000 | 1.285300 | 0.000000 | 0.236  | 0.001 | *** |
| V7WEHE   | 0.013287 | 0.005547 | 2.395000 | 1.252500 | 0.000000 | 0.255  | 0.001 | *** |
| V60LFB   | 0.013139 | 0.004117 | 3.192000 | 0.076800 | 1.312300 | 0.275  | 0.001 | *** |
| V21ESP   | 0.013118 | 0.006422 | 2.043000 | 1.236300 | 0.000000 | 0.294  | 0.001 | *** |
| V25WAG   | 0.012318 | 0.004959 | 2.484000 | 0.205100 | 1.365200 | 0.312  | 0.001 | *** |
| V31AMAG  | 0.011976 | 0.005536 | 2.163000 | 0.370900 | 1.393900 | 0.330  | 0.001 | *** |
| V45MGLK  | 0.011411 | 0.003990 | 2.860000 | 0.142600 | 1.216900 | 0.347  | 0.001 | *** |
| V36YFHE  | 0.011370 | 0.007879 | 1.443000 | 1.121400 | 0.000000 | 0.364  | 0.043 | *   |
| V13RWB   | 0.011326 | 0.007442 | 1.522000 | 1.080500 | 0.000000 | 0.381  | 0.059 | .   |
| V47RFC   | 0.011238 | 0.004799 | 2.342000 | 0.214300 | 1.251400 | 0.397  | 0.001 | *** |
| V70RSL   | 0.011018 | 0.007241 | 1.522000 | 0.094600 | 1.085200 | 0.414  | 0.003 | **  |
| V41SPP   | 0.010892 | 0.006237 | 1.746000 | 1.058800 | 0.000000 | 0.430  | 0.004 | **  |
| V15STTH  | 0.010499 | 0.008202 | 1.280000 | 1.475400 | 0.745400 | 0.445  | 0.120 |     |
| V35STP   | 0.010090 | 0.007467 | 1.351000 | 0.700100 | 1.638300 | 0.460  | 0.030 | *   |
| V73JW    | 0.009687 | 0.005914 | 1.638000 | 0.000000 | 0.934100 | 0.474  | 0.001 | *** |
| V33RWH   | 0.009674 | 0.006163 | 1.570000 | 1.318700 | 0.532700 | 0.489  | 0.030 | *   |
| V42SIL   | 0.009667 | 0.007712 | 1.253000 | 0.952800 | 0.442300 | 0.503  | 0.112 |     |
| V34WSW   | 0.009358 | 0.006570 | 1.424000 | 0.247500 | 0.961700 | 0.517  | 0.117 |     |
| V52TRM   | 0.009124 | 0.005824 | 1.567000 | 0.202800 | 0.871300 | 0.531  | 0.009 | **  |
| V98FLAME | 0.009079 | 0.006396 | 1.419000 | 0.220500 | 0.964700 | 0.544  | 0.008 | **  |
| V69SKF   | 0.008992 | 0.005367 | 1.675000 | 0.334500 | 1.156400 | 0.557  | 0.006 | **  |
| V18YTH   | 0.008930 | 0.008900 | 1.003000 | 0.171800 | 0.845000 | 0.571  | 0.286 |     |
| V23RBFT  | 0.008751 | 0.006449 | 1.357000 | 0.990600 | 0.456500 | 0.584  | 0.373 |     |
| V6WTTT   | 0.008747 | 0.007272 | 1.203000 | 1.325800 | 0.736100 | 0.597  | 0.172 |     |
| V56FTC   | 0.008722 | 0.005609 | 1.555000 | 0.948600 | 0.220000 | 0.609  | 0.023 | *   |
| V93DB    | 0.008385 | 0.005276 | 1.589000 | 0.061600 | 0.829800 | 0.622  | 0.001 | *** |
| V91NMIN  | 0.008172 | 0.007902 | 1.034000 | 0.103300 | 0.767400 | 0.634  | 0.026 | *   |
| V28VS    | 0.007982 | 0.007161 | 1.115000 | 0.599300 | 0.686000 | 0.646  | 0.624 |     |
| V56GWH   | 0.007767 | 0.005012 | 1.550000 | 1.325200 | 0.765000 | 0.657  | 0.077 | .   |
| V94RBEE  | 0.007444 | 0.004665 | 1.596000 | 0.053800 | 0.732200 | 0.668  | 0.001 | *** |
| V27NHHE  | 0.007397 | 0.008745 | 0.846000 | 0.689200 | 0.000000 | 0.679  | 0.676 |     |
| V19ER    | 0.007375 | 0.006436 | 1.146000 | 0.816000 | 1.397300 | 0.690  | 0.647 |     |
| V16LR    | 0.006856 | 0.005658 | 1.212000 | 0.852900 | 1.252000 | 0.700  | 0.880 |     |
| V59YR    | 0.006854 | 0.006952 | 0.986000 | 0.000000 | 0.651800 | 0.711  | 0.011 | *   |
| V81SHBC  | 0.006705 | 0.005721 | 1.172000 | 0.641600 | 0.000000 | 0.720  | 0.293 |     |
| V36F     | 0.006585 | 0.003090 | 2.131000 | 1.569300 | 1.077300 | 0.730  | 0.278 |     |
| V29CST   | 0.006553 | 0.005132 | 1.277000 | 0.548200 | 0.776000 | 0.740  | 0.652 |     |

|          |          |          |          |          |          |       |          |
|----------|----------|----------|----------|----------|----------|-------|----------|
| V20PCU   | 0.006505 | 0.006055 | 1.074000 | 0.411800 | 0.602800 | 0.750 | 0.551    |
| V88WG    | 0.006448 | 0.006586 | 0.979000 | 0.000000 | 0.612900 | 0.759 | 0.011 *  |
| V90WTG   | 0.006446 | 0.006479 | 0.995000 | 0.055600 | 0.612900 | 0.769 | 0.028 *  |
| V99WWT   | 0.006439 | 0.006551 | 0.983000 | 0.000000 | 0.615300 | 0.778 | 0.008 ** |
| V580B0   | 0.006407 | 0.006318 | 1.014000 | 0.198800 | 0.611700 | 0.788 | 0.105    |
| V24BFCS  | 0.006242 | 0.005422 | 1.151000 | 0.751600 | 1.328700 | 0.797 | 0.209    |
| V9SFW    | 0.006159 | 0.003969 | 1.552000 | 1.384400 | 1.145500 | 0.806 | 0.352    |
| V80RF    | 0.005971 | 0.006465 | 0.924000 | 0.568000 | 0.000000 | 0.815 | 0.501    |
| V50CHE   | 0.005950 | 0.007394 | 0.805000 | 0.580200 | 0.000000 | 0.824 | 0.531    |
| V92NFB   | 0.005676 | 0.005583 | 1.017000 | 0.141400 | 0.513100 | 0.832 | 0.019 *  |
| V72CRP   | 0.005564 | 0.005644 | 0.986000 | 0.000000 | 0.545600 | 0.840 | 0.004 ** |
| V71PD0V  | 0.005528 | 0.005717 | 0.967000 | 0.055600 | 0.530500 | 0.849 | 0.015 *  |
| V48YTBC  | 0.005233 | 0.006402 | 0.817000 | 0.511500 | 0.000000 | 0.856 | 0.569    |
| V14AUR   | 0.005141 | 0.005536 | 0.929000 | 0.893600 | 1.237700 | 0.864 | 0.753    |
| V12LK    | 0.005031 | 0.005700 | 0.883000 | 1.372500 | 0.910300 | 0.871 | 0.120    |
| V46BHHE  | 0.004971 | 0.006225 | 0.799000 | 0.491200 | 0.000000 | 0.879 | 0.997    |
| V37WHIP  | 0.004588 | 0.006321 | 0.726000 | 0.441900 | 0.000000 | 0.885 | 0.672    |
| V84YRTH  | 0.004512 | 0.007311 | 0.617000 | 0.082700 | 0.389600 | 0.892 | 0.027 *  |
| V83SFC   | 0.004484 | 0.006299 | 0.712000 | 0.424600 | 0.000000 | 0.899 | 0.817    |
| V43GCU   | 0.004425 | 0.006145 | 0.720000 | 0.423100 | 0.000000 | 0.905 | 0.986    |
| V87LFC   | 0.004386 | 0.006114 | 0.717000 | 0.224900 | 0.314400 | 0.912 | 0.203    |
| V53MB    | 0.003834 | 0.005402 | 0.710000 | 0.224200 | 0.210200 | 0.917 | 0.660    |
| V101LCOR | 0.003699 | 0.006477 | 0.571000 | 0.000000 | 0.370000 | 0.923 | 0.064 .  |
| V55LHE   | 0.003608 | 0.005688 | 0.634000 | 0.354900 | 0.000000 | 0.928 | 0.624    |
| V63DWS   | 0.003160 | 0.005040 | 0.627000 | 0.313100 | 0.000000 | 0.933 | 0.833    |
| V65LWB   | 0.002918 | 0.008100 | 0.360000 | 0.258700 | 0.000000 | 0.937 | 0.603    |
| V49LYRE  | 0.002876 | 0.005277 | 0.545000 | 0.268000 | 0.000000 | 0.942 | 0.704    |
| V74BCHE  | 0.002855 | 0.005003 | 0.571000 | 0.000000 | 0.261700 | 0.946 | 0.052 .  |
| V85ROSE  | 0.002515 | 0.004606 | 0.546000 | 0.239700 | 0.000000 | 0.950 | 0.536    |
| V39FHE   | 0.002455 | 0.006886 | 0.357000 | 0.248500 | 0.000000 | 0.953 | 0.865    |
| V68PIL   | 0.002403 | 0.005266 | 0.456000 | 0.232900 | 0.000000 | 0.957 | 0.716    |
| V44MUSK  | 0.002307 | 0.006375 | 0.362000 | 0.236300 | 0.000000 | 0.960 | 0.769    |
| V95HBC   | 0.002170 | 0.003801 | 0.571000 | 0.000000 | 0.198800 | 0.963 | 0.168    |
| V510WH   | 0.002158 | 0.004757 | 0.454000 | 0.211500 | 0.000000 | 0.967 | 0.458    |
| V22SCR   | 0.002084 | 0.004592 | 0.454000 | 0.210400 | 0.000000 | 0.970 | 0.999    |
| V16ST    | 0.002006 | 0.001368 | 1.466000 | 1.466600 | 1.343300 | 0.973 | 0.623    |
| V77RRP   | 0.001709 | 0.004743 | 0.360000 | 0.174100 | 0.000000 | 0.975 | 0.346    |
| V76GBB   | 0.001680 | 0.003676 | 0.457000 | 0.166700 | 0.000000 | 0.978 | 0.655    |
| V86BC00  | 0.001432 | 0.003963 | 0.361000 | 0.142300 | 0.000000 | 0.980 | 0.625    |
| V67GGC   | 0.001395 | 0.003914 | 0.356000 | 0.135400 | 0.000000 | 0.982 | 0.878    |
| V64BELL  | 0.001382 | 0.005570 | 0.248000 | 0.127500 | 0.000000 | 0.984 | 0.570    |
| V62RBTR  | 0.001356 | 0.003781 | 0.359000 | 0.127400 | 0.000000 | 0.986 | 0.684    |
| V57PINK  | 0.001324 | 0.003673 | 0.360000 | 0.127300 | 0.000000 | 0.988 | 0.492    |
| V97PCL   | 0.001101 | 0.004441 | 0.248000 | 0.102200 | 0.000000 | 0.990 | 0.404    |
| V40BRTH  | 0.001017 | 0.004100 | 0.248000 | 0.088800 | 0.000000 | 0.991 | 0.997    |
| V78LLOR  | 0.000946 | 0.003812 | 0.248000 | 0.106900 | 0.000000 | 0.992 | 0.458    |
| V89PC00  | 0.000744 | 0.003002 | 0.248000 | 0.069100 | 0.000000 | 0.994 | 0.404    |
| V54STHR  | 0.000693 | 0.002793 | 0.248000 | 0.073200 | 0.000000 | 0.995 | 0.859    |
| V79YTHE  | 0.000667 | 0.002689 | 0.248000 | 0.075400 | 0.000000 | 0.996 | 0.776    |
| V66CBW   | 0.000652 | 0.002628 | 0.248000 | 0.058800 | 0.000000 | 0.997 | 0.930    |
| V96DF    | 0.000626 | 0.002525 | 0.248000 | 0.070800 | 0.000000 | 0.997 | 0.810    |
| V82AZKF  | 0.000602 | 0.002428 | 0.248000 | 0.053800 | 0.000000 | 0.998 | 0.433    |
| V100WBWS | 0.000585 | 0.002359 | 0.248000 | 0.066200 | 0.000000 | 0.999 | 0.458    |
| V102KING | 0.000545 | 0.002198 | 0.248000 | 0.053800 | 0.000000 | 1.000 | 0.626    |
| V61SPW   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA       |
| V75RCR   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA       |

Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Contrast: Gippsland Ma\_Montane Fore

|         | average  | sd       | ratio    | ava      | avb      | cumsum | p         |
|---------|----------|----------|----------|----------|----------|--------|-----------|
| V17WPHE | 0.025511 | 0.002976 | 8.572000 | 2.247700 | 0.000000 | 0.047  | 0.001 *** |
| V34WSW  | 0.020923 | 0.002668 | 7.841000 | 1.856000 | 0.000000 | 0.086  | 0.002 **  |
| V11CR   | 0.019365 | 0.002542 | 7.617000 | 0.000000 | 1.710400 | 0.121  | 0.002 **  |

|          |          |          |           |          |          |       |       |     |
|----------|----------|----------|-----------|----------|----------|-------|-------|-----|
| V27NHHE  | 0.018629 | 0.005197 | 3.584000  | 1.646300 | 0.000000 | 0.156 | 0.001 | *** |
| V6WTRR   | 0.017764 | 0.002227 | 7.977000  | 0.000000 | 1.567100 | 0.189 | 0.001 | *** |
| V15STTH  | 0.017530 | 0.007233 | 2.423000  | 0.323400 | 1.857500 | 0.221 | 0.001 | *** |
| V25WAG   | 0.015594 | 0.002143 | 7.276000  | 1.376000 | 0.000000 | 0.250 | 0.001 | *** |
| V13RWB   | 0.014804 | 0.009495 | 1.559000  | 1.922100 | 0.662400 | 0.277 | 0.014 | *   |
| V20PCU   | 0.014406 | 0.001861 | 7.739000  | 0.000000 | 1.268100 | 0.304 | 0.001 | *** |
| V43GCU   | 0.014108 | 0.001624 | 8.689000  | 0.000000 | 1.256100 | 0.330 | 0.002 | **  |
| V83SFC   | 0.013763 | 0.001275 | 10.794000 | 0.000000 | 1.220000 | 0.355 | 0.001 | *** |
| V4BTH    | 0.012989 | 0.009916 | 1.310000  | 1.184200 | 2.294000 | 0.379 | 0.080 | .   |
| V42SIL   | 0.012586 | 0.005694 | 2.210000  | 0.420400 | 1.363600 | 0.402 | 0.035 | *   |
| V36F     | 0.012521 | 0.008426 | 1.486000  | 0.646500 | 1.724600 | 0.425 | 0.004 | **  |
| V19ER    | 0.011718 | 0.005658 | 2.071000  | 1.308600 | 0.266900 | 0.447 | 0.019 | *   |
| V37WHIP  | 0.011155 | 0.006729 | 1.658000  | 0.000000 | 1.020400 | 0.468 | 0.007 | **  |
| V48YTBC  | 0.010991 | 0.006611 | 1.663000  | 0.000000 | 0.958400 | 0.488 | 0.006 | **  |
| V81SHBC  | 0.010400 | 0.005715 | 1.834000  | 0.271900 | 1.191300 | 0.507 | 0.007 | **  |
| V21ESP   | 0.010274 | 0.006547 | 1.569000  | 0.644000 | 1.502100 | 0.526 | 0.139 | .   |
| V67GGC   | 0.010101 | 0.006264 | 1.613000  | 0.000000 | 0.926600 | 0.545 | 0.002 | **  |
| V80RF    | 0.010024 | 0.006191 | 1.619000  | 0.000000 | 0.913500 | 0.563 | 0.019 | *   |
| V56WH    | 0.009874 | 0.007909 | 1.248000  | 0.674800 | 1.486100 | 0.582 | 0.032 | *   |
| V29CST   | 0.009751 | 0.004639 | 2.102000  | 1.059100 | 0.314400 | 0.599 | 0.031 | *   |
| V36YFHE  | 0.009681 | 0.007030 | 1.377000  | 1.243700 | 1.134500 | 0.617 | 0.425 | .   |
| V16LR    | 0.009375 | 0.007159 | 1.310000  | 1.184900 | 1.023200 | 0.635 | 0.232 | .   |
| V46BHHHE | 0.008981 | 0.008370 | 1.073000  | 0.307900 | 0.828800 | 0.651 | 0.375 | .   |
| V49LYRE  | 0.008738 | 0.005375 | 1.626000  | 0.000000 | 0.803100 | 0.667 | 0.012 | *   |
| V33RWH   | 0.008651 | 0.007180 | 1.205000  | 0.579800 | 0.753700 | 0.683 | 0.269 | .   |
| V68PIL   | 0.008604 | 0.005317 | 1.618000  | 0.000000 | 0.785200 | 0.699 | 0.011 | *   |
| V54STHR  | 0.008499 | 0.005150 | 1.650000  | 0.000000 | 0.758900 | 0.715 | 0.001 | *** |
| V65LWB   | 0.008071 | 0.008400 | 0.961000  | 0.741500 | 0.000000 | 0.730 | 0.151 | .   |
| V63DWS   | 0.007899 | 0.008217 | 0.961000  | 0.724800 | 0.000000 | 0.744 | 0.077 | .   |
| V91NMIN  | 0.007851 | 0.008209 | 0.956000  | 0.719700 | 0.000000 | 0.759 | 0.080 | .   |
| V28VS    | 0.007829 | 0.008154 | 0.960000  | 0.000000 | 0.709200 | 0.773 | 0.598 | .   |
| V7WEHE   | 0.007421 | 0.005457 | 1.360000  | 1.241500 | 1.224500 | 0.787 | 0.402 | .   |
| V24BFCS  | 0.007277 | 0.005961 | 1.221000  | 1.112500 | 0.496700 | 0.800 | 0.103 | .   |
| V56FTC   | 0.007098 | 0.006534 | 1.086000  | 0.615700 | 1.231300 | 0.813 | 0.411 | .   |
| V35STP   | 0.006718 | 0.007148 | 0.940000  | 0.993200 | 0.944200 | 0.826 | 0.815 | .   |
| V23RBFT  | 0.006640 | 0.006306 | 1.053000  | 0.937700 | 0.883700 | 0.838 | 0.873 | .   |
| V14AUR   | 0.006610 | 0.006785 | 0.974000  | 0.962100 | 0.919800 | 0.850 | 0.417 | .   |
| V55LHE   | 0.006240 | 0.006468 | 0.965000  | 0.000000 | 0.590500 | 0.862 | 0.157 | .   |
| V12LK    | 0.005918 | 0.005486 | 1.079000  | 0.801000 | 1.324100 | 0.873 | 0.077 | .   |
| V26WWCH  | 0.005801 | 0.010413 | 0.557000  | 0.000000 | 0.462400 | 0.883 | 0.885 | .   |
| V50CHE   | 0.005678 | 0.005941 | 0.956000  | 0.000000 | 0.509600 | 0.894 | 0.590 | .   |
| V86BC00  | 0.005379 | 0.005588 | 0.962000  | 0.000000 | 0.481700 | 0.904 | 0.069 | .   |
| V22SCR   | 0.003983 | 0.007149 | 0.557000  | 0.000000 | 0.317500 | 0.911 | 0.835 | .   |
| V8WNHE   | 0.003758 | 0.001583 | 2.374000  | 1.843200 | 1.709100 | 0.918 | 0.863 | .   |
| V85ROSE  | 0.003564 | 0.006391 | 0.558000  | 0.000000 | 0.331700 | 0.925 | 0.285 | .   |
| V62RBTR  | 0.003528 | 0.006332 | 0.557000  | 0.000000 | 0.281200 | 0.931 | 0.192 | .   |
| V38GAL   | 0.003497 | 0.006277 | 0.557000  | 0.323400 | 0.000000 | 0.938 | 0.828 | .   |
| V87LFC   | 0.003183 | 0.005712 | 0.557000  | 0.289600 | 0.000000 | 0.944 | 0.566 | .   |
| V45MGLK  | 0.002941 | 0.005278 | 0.557000  | 0.271900 | 0.000000 | 0.949 | 0.818 | .   |
| V41SPP   | 0.002919 | 0.001917 | 1.523000  | 1.200500 | 1.352200 | 0.954 | 0.897 | .   |
| V9SFW    | 0.002819 | 0.002623 | 1.075000  | 1.734700 | 1.525000 | 0.960 | 0.906 | .   |
| V53MB    | 0.002704 | 0.004852 | 0.557000  | 0.250000 | 0.000000 | 0.965 | 0.767 | .   |
| V57PINK  | 0.002686 | 0.004816 | 0.558000  | 0.000000 | 0.250000 | 0.970 | 0.265 | .   |
| V76GBB   | 0.002568 | 0.004606 | 0.557000  | 0.000000 | 0.220000 | 0.974 | 0.395 | .   |
| V70RSL   | 0.002557 | 0.004589 | 0.557000  | 0.236400 | 0.000000 | 0.979 | 0.835 | .   |
| V2EYR    | 0.002508 | 0.001941 | 1.292000  | 1.599500 | 1.440100 | 0.984 | 0.914 | .   |
| V102KING | 0.002459 | 0.004408 | 0.558000  | 0.000000 | 0.236400 | 0.988 | 0.078 | .   |
| V51OWH   | 0.002364 | 0.004239 | 0.558000  | 0.000000 | 0.220000 | 0.993 | 0.528 | .   |
| V10WBSW  | 0.002324 | 0.001192 | 1.950000  | 1.760400 | 1.622600 | 0.997 | 0.984 | .   |
| V16ST    | 0.001754 | 0.001373 | 1.278000  | 1.493300 | 1.379200 | 1.000 | 0.762 | .   |
| V18YTH   | 0.000000 | 0.000000 | NaN       | 0.000000 | 0.000000 | 1.000 | NA    | .   |
| V30BTR   | 0.000000 | 0.000000 | NaN       | 0.000000 | 0.000000 | 1.000 | NA    | .   |
| V31AMAG  | 0.000000 | 0.000000 | NaN       | 0.000000 | 0.000000 | 1.000 | NA    | .   |
| V32SCC   | 0.000000 | 0.000000 | NaN       | 0.000000 | 0.000000 | 1.000 | NA    | .   |
| V39FHE   | 0.000000 | 0.000000 | NaN       | 0.000000 | 0.000000 | 1.000 | NA    | .   |

|          |          |          |     |          |          |       |    |
|----------|----------|----------|-----|----------|----------|-------|----|
| V40BRTH  | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V44MUSK  | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V47RFC   | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V52TRM   | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V580B0   | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V59YR    | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V60LFB   | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V61SPW   | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V64BELL  | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V66CBW   | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V69SKF   | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V71PD0V  | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V72CRP   | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V73JW    | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V74BCHE  | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V75RCR   | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V77RRP   | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V78LL0R  | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V79YTHE  | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V82AZKF  | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V84YRTH  | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V88WG    | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V89PC00  | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V90WTG   | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V92NFB   | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V93DB    | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V94RBEE  | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V95HBC   | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V96DF    | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V97PCL   | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V98FLAME | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V99WWT   | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V100WBWS | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V101LCOR | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |

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Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Contrast: Gippsland Ma\_Foothills Wo

|         | average  | sd       | ratio     | ava      | avb      | cumsum | p         |
|---------|----------|----------|-----------|----------|----------|--------|-----------|
| V17WPHE | 0.026097 | 0.002383 | 10.951000 | 2.247700 | 0.000000 | 0.052  | 0.001 *** |
| V34WSW  | 0.021402 | 0.002230 | 9.597000  | 1.856000 | 0.000000 | 0.094  | 0.001 *** |
| V6WTTR  | 0.018324 | 0.001427 | 12.839000 | 0.000000 | 1.579900 | 0.130  | 0.001 *** |
| V11CR   | 0.017226 | 0.001813 | 9.499000  | 0.000000 | 1.484400 | 0.164  | 0.005 **  |
| V15STTH | 0.017112 | 0.007416 | 2.307000  | 0.323400 | 1.788200 | 0.198  | 0.003 **  |
| V40BRTH | 0.016327 | 0.010384 | 1.572000  | 0.000000 | 1.423800 | 0.230  | 0.001 *** |
| V27NHHE | 0.015988 | 0.007215 | 2.216000  | 1.646300 | 0.256000 | 0.262  | 0.011 *   |
| V25WAG  | 0.015952 | 0.001857 | 8.589000  | 1.376000 | 0.000000 | 0.293  | 0.001 *** |
| V22SCR  | 0.015801 | 0.001670 | 9.460000  | 0.000000 | 1.364500 | 0.325  | 0.001 *** |
| V26WWCH | 0.013884 | 0.008667 | 1.602000  | 0.000000 | 1.210300 | 0.352  | 0.022 *   |
| V18YTH  | 0.013759 | 0.008550 | 1.609000  | 0.000000 | 1.199300 | 0.379  | 0.022 *   |
| V28VS   | 0.012735 | 0.007685 | 1.657000  | 0.000000 | 1.088900 | 0.404  | 0.020 *   |
| V10WBWS | 0.011739 | 0.009043 | 1.298000  | 1.760400 | 0.773600 | 0.428  | 0.093 .   |
| V36F    | 0.011589 | 0.007893 | 1.468000  | 0.646500 | 1.569600 | 0.451  | 0.006 **  |
| V33RWH  | 0.011462 | 0.006709 | 1.709000  | 0.579800 | 1.597500 | 0.473  | 0.027 *   |
| V69SKF  | 0.010219 | 0.006162 | 1.658000  | 0.000000 | 0.890600 | 0.494  | 0.008 **  |
| V4BTH   | 0.010179 | 0.009715 | 1.048000  | 1.184200 | 1.975000 | 0.514  | 0.350     |
| V23RBT  | 0.009465 | 0.006668 | 1.419000  | 0.937700 | 0.243500 | 0.532  | 0.293     |
| V46BHHE | 0.009439 | 0.007230 | 1.305000  | 0.307900 | 0.931700 | 0.551  | 0.279     |
| V42SIL  | 0.009295 | 0.005353 | 1.736000  | 0.420400 | 0.761200 | 0.569  | 0.286     |
| V21ESP  | 0.009082 | 0.006693 | 1.357000  | 0.644000 | 1.346100 | 0.587  | 0.334     |
| V56WH   | 0.008800 | 0.007323 | 1.202000  | 0.674800 | 1.309900 | 0.605  | 0.088 .   |
| V13RWB  | 0.008699 | 0.002320 | 3.750000  | 1.922100 | 1.170600 | 0.622  | 0.647     |
| V53MB   | 0.008525 | 0.006365 | 1.339000  | 0.250000 | 0.851600 | 0.639  | 0.029 *   |
| V36YFHE | 0.008313 | 0.004873 | 1.706000  | 1.243700 | 1.319700 | 0.655  | 0.721     |

|         |          |          |          |          |          |       |         |
|---------|----------|----------|----------|----------|----------|-------|---------|
| V65LWB  | 0.008250 | 0.008548 | 0.965000 | 0.741500 | 0.000000 | 0.671 | 0.142   |
| V63DWS  | 0.008179 | 0.007487 | 1.092000 | 0.724800 | 0.243500 | 0.688 | 0.054   |
| V81SHBC | 0.008171 | 0.006178 | 1.323000 | 0.271900 | 0.807100 | 0.704 | 0.103   |
| V19ER   | 0.008140 | 0.007125 | 1.143000 | 1.308600 | 0.654600 | 0.720 | 0.353   |
| V91NMIN | 0.008026 | 0.008354 | 0.961000 | 0.719700 | 0.000000 | 0.736 | 0.074   |
| V7WEHE  | 0.007740 | 0.006507 | 1.189000 | 1.241500 | 1.318200 | 0.751 | 0.332   |
| V29CST  | 0.007736 | 0.004933 | 1.568000 | 1.059100 | 0.659900 | 0.766 | 0.183   |
| V20PCU  | 0.007716 | 0.008005 | 0.964000 | 0.000000 | 0.673000 | 0.782 | 0.186   |
| V9SFW   | 0.007643 | 0.007705 | 0.992000 | 1.734700 | 1.081900 | 0.797 | 0.223   |
| V56FTC  | 0.007448 | 0.006647 | 1.120000 | 0.615700 | 1.244700 | 0.811 | 0.293   |
| V14AUR  | 0.006977 | 0.007154 | 0.975000 | 0.962100 | 0.987000 | 0.825 | 0.389   |
| V35STP  | 0.006914 | 0.007551 | 0.916000 | 0.993200 | 1.011200 | 0.839 | 0.780   |
| V43GCU  | 0.006636 | 0.007051 | 0.941000 | 0.000000 | 0.583000 | 0.852 | 0.530   |
| V12LK   | 0.006350 | 0.005685 | 1.117000 | 0.801000 | 1.350100 | 0.865 | 0.040 * |
| V16LR   | 0.006246 | 0.006500 | 0.961000 | 1.184900 | 1.479800 | 0.877 | 0.823   |
| V64BELL | 0.005568 | 0.009988 | 0.557000 | 0.000000 | 0.492000 | 0.888 | 0.226   |
| V50CHE  | 0.004608 | 0.008269 | 0.557000 | 0.000000 | 0.384600 | 0.897 | 0.698   |
| V96DF   | 0.004006 | 0.007186 | 0.557000 | 0.000000 | 0.349000 | 0.905 | 0.099   |
| V41SPP  | 0.003877 | 0.002906 | 1.334000 | 1.200500 | 1.526000 | 0.913 | 0.902   |
| V24BFCS | 0.003777 | 0.005287 | 0.714000 | 1.112500 | 0.794400 | 0.920 | 0.858   |
| V61SPW  | 0.003744 | 0.006716 | 0.557000 | 0.000000 | 0.326200 | 0.927 | 0.108   |
| V80RF   | 0.003648 | 0.006547 | 0.557000 | 0.000000 | 0.304500 | 0.935 | 0.854   |
| V2EYR   | 0.003593 | 0.002576 | 1.394000 | 1.599500 | 1.316800 | 0.942 | 0.866   |
| V38GAL  | 0.003574 | 0.006396 | 0.559000 | 0.323400 | 0.000000 | 0.949 | 0.795   |
| V87LFC  | 0.003254 | 0.005822 | 0.559000 | 0.289600 | 0.000000 | 0.955 | 0.535   |
| V8WNHE  | 0.003139 | 0.002678 | 1.172000 | 1.843200 | 1.582200 | 0.962 | 0.915   |
| V580B0  | 0.003041 | 0.005456 | 0.557000 | 0.000000 | 0.261700 | 0.968 | 0.796   |
| V83SFC  | 0.003041 | 0.005456 | 0.557000 | 0.000000 | 0.261700 | 0.974 | 0.888   |
| V45MGLK | 0.003006 | 0.005378 | 0.559000 | 0.271900 | 0.000000 | 0.980 | 0.796   |
| V16ST   | 0.002663 | 0.001997 | 1.334000 | 1.493300 | 1.577800 | 0.985 | 0.207   |
| V70RSL  | 0.002613 | 0.004676 | 0.559000 | 0.236400 | 0.000000 | 0.990 | 0.833   |
| V62RBTR | 0.002557 | 0.004588 | 0.557000 | 0.000000 | 0.220000 | 0.995 | 0.450   |
| V66CBW  | 0.002557 | 0.004588 | 0.557000 | 0.000000 | 0.220000 | 1.000 | 0.523   |
| V30BTR  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |
| V31AMAG | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |
| V32SCC  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |
| V37WHIP | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |
| V39FHE  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |
| V44MUSK | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |
| V47RFC  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |
| V48YTBC | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |
| V49LYRE | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |
| V51OWH  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |
| V52TRM  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |
| V54STHR | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |
| V55LHE  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |
| V57PINK | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |
| V59YR   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |
| V60LFB  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |
| V67GGC  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |
| V68PIL  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |
| V71PD0V | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |
| V72CRP  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |
| V73JW   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |
| V74BCHE | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |
| V75RCR  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |
| V76GBB  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |
| V77RRP  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |
| V78LLOR | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |
| V79YTHE | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |
| V82AZKF | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |
| V84YRTH | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |
| V85ROSE | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |
| V86BC00 | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |
| V88WG   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |



|          |          |          |     |          |          |       |    |
|----------|----------|----------|-----|----------|----------|-------|----|
| V89PC00  | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V90WTG   | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V92NFB   | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V93DB    | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V94RBEE  | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V95HBC   | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V97PCL   | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V98FLAME | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V99WWT   | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V100WBWS | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V101LCOR | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V102KING | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |

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Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Contrast: Gippsland Ma\_Box-Ironbark

|         | average  | sd       | ratio    | ava      | avb      | cumsum      | p   |
|---------|----------|----------|----------|----------|----------|-------------|-----|
| V17WPHE | 0.026261 | 0.003224 | 8.146000 | 2.247700 | 0.000000 | 0.046 0.001 | *** |
| V18YTH  | 0.022239 | 0.002654 | 8.378000 | 0.000000 | 1.909900 | 0.086 0.001 | *** |
| V34WSW  | 0.021534 | 0.002823 | 7.627000 | 1.856000 | 0.000000 | 0.124 0.001 | *** |
| V10WBWS | 0.020600 | 0.003335 | 6.178000 | 1.760400 | 0.000000 | 0.160 0.001 | *** |
| V26WWCH | 0.019377 | 0.002919 | 6.638000 | 0.000000 | 1.655500 | 0.195 0.001 | *** |
| V27NHHE | 0.019175 | 0.005389 | 3.559000 | 1.646300 | 0.000000 | 0.229 0.002 | **  |
| V6WTTT  | 0.017796 | 0.001927 | 9.234000 | 0.000000 | 1.525100 | 0.260 0.001 | *** |
| V46BHHE | 0.017556 | 0.007155 | 2.454000 | 0.307900 | 1.798900 | 0.291 0.001 | *** |
| V39FHE  | 0.017550 | 0.003257 | 5.388000 | 0.000000 | 1.532700 | 0.322 0.001 | *** |
| V25WAG  | 0.016052 | 0.002282 | 7.033000 | 1.376000 | 0.000000 | 0.351 0.001 | *** |
| V22SCR  | 0.015788 | 0.003354 | 4.707000 | 0.000000 | 1.350800 | 0.379 0.001 | *** |
| V40BRTT | 0.015511 | 0.009337 | 1.661000 | 0.000000 | 1.379300 | 0.406 0.003 | **  |
| V15STTH | 0.015362 | 0.007523 | 2.042000 | 0.323400 | 1.627000 | 0.433 0.007 | **  |
| V44MUSK | 0.013318 | 0.008639 | 1.542000 | 0.000000 | 1.180800 | 0.457 0.006 | **  |
| V580B0  | 0.012439 | 0.003104 | 4.007000 | 0.000000 | 1.059200 | 0.479 0.001 | *** |
| V42SIL  | 0.012248 | 0.004880 | 2.510000 | 0.420400 | 1.255000 | 0.500 0.037 | *   |
| V75RCR  | 0.011824 | 0.003514 | 3.365000 | 0.000000 | 1.000900 | 0.521 0.001 | *** |
| V28VS   | 0.011169 | 0.006772 | 1.649000 | 0.000000 | 0.971800 | 0.541 0.064 | .   |
| V11CR   | 0.011098 | 0.006986 | 1.589000 | 0.000000 | 0.992900 | 0.561 0.166 | .   |
| V33RWH  | 0.010578 | 0.006786 | 1.559000 | 0.579800 | 1.512200 | 0.580 0.037 | *   |
| V16LR   | 0.010525 | 0.007973 | 1.320000 | 1.184900 | 0.576500 | 0.598 0.096 | .   |
| V36F    | 0.010144 | 0.006630 | 1.530000 | 0.646500 | 1.330200 | 0.616 0.048 | *   |
| V29CST  | 0.009753 | 0.005485 | 1.778000 | 1.059100 | 0.256000 | 0.633 0.036 | *   |
| V23RBFT | 0.009549 | 0.007039 | 1.357000 | 0.937700 | 0.320500 | 0.650 0.293 | .   |
| V43GCU  | 0.009226 | 0.005611 | 1.644000 | 0.000000 | 0.818400 | 0.667 0.053 | .   |
| V35STP  | 0.009032 | 0.007151 | 1.263000 | 0.993200 | 0.656700 | 0.683 0.249 | .   |
| V36YFHE | 0.008694 | 0.006441 | 1.350000 | 1.243700 | 1.383500 | 0.698 0.640 | .   |
| V2EYR   | 0.008634 | 0.006941 | 1.244000 | 1.599500 | 0.874900 | 0.713 0.168 | .   |
| V56WH   | 0.008626 | 0.006596 | 1.308000 | 0.674800 | 1.171900 | 0.729 0.118 | .   |
| V65LWB  | 0.008297 | 0.008644 | 0.960000 | 0.741500 | 0.000000 | 0.743 0.148 | .   |
| V63DWS  | 0.008199 | 0.007646 | 1.072000 | 0.724800 | 0.220000 | 0.758 0.047 | *   |
| V91NMIN | 0.008072 | 0.008448 | 0.955000 | 0.719700 | 0.000000 | 0.772 0.079 | .   |
| V21ESP  | 0.007818 | 0.008323 | 0.939000 | 0.644000 | 0.000000 | 0.786 0.713 | .   |
| V7WEHE  | 0.007692 | 0.006430 | 1.196000 | 1.241500 | 1.296700 | 0.799 0.338 | .   |
| V4BTH   | 0.007495 | 0.007712 | 0.972000 | 1.184200 | 1.520900 | 0.813 0.747 | .   |
| V56FTC  | 0.007373 | 0.007510 | 0.982000 | 0.615700 | 0.281200 | 0.826 0.339 | .   |
| V66CBW  | 0.007079 | 0.007358 | 0.962000 | 0.000000 | 0.566600 | 0.838 0.004 | **  |
| V90WTG  | 0.006660 | 0.007437 | 0.896000 | 0.000000 | 0.542000 | 0.850 0.037 | *   |
| V70RSL  | 0.006539 | 0.006415 | 1.019000 | 0.236400 | 0.557400 | 0.862 0.256 | .   |
| V61SPW  | 0.006393 | 0.006678 | 0.957000 | 0.000000 | 0.593000 | 0.873 0.021 | *   |
| V19ER   | 0.006130 | 0.006577 | 0.932000 | 1.308600 | 0.846400 | 0.884 0.816 | .   |
| V14AUR  | 0.005432 | 0.006537 | 0.831000 | 0.962100 | 1.211600 | 0.893 0.657 | .   |
| V81SHBC | 0.004610 | 0.006122 | 0.753000 | 0.271900 | 0.281200 | 0.902 0.932 | .   |
| V12LK   | 0.004495 | 0.005039 | 0.892000 | 0.801000 | 1.122800 | 0.910 0.304 | .   |
| V79YTHE | 0.004087 | 0.007327 | 0.558000 | 0.000000 | 0.396100 | 0.917 0.040 | *   |
| V9SFW   | 0.004012 | 0.003799 | 1.056000 | 1.734700 | 1.472800 | 0.924 0.729 | .   |
| V30BTR  | 0.003950 | 0.007083 | 0.558000 | 0.000000 | 0.382900 | 0.931 0.641 | .   |

|          |          |          |          |          |          |       |         |
|----------|----------|----------|----------|----------|----------|-------|---------|
| V38GAL   | 0.003595 | 0.006455 | 0.557000 | 0.323400 | 0.000000 | 0.937 | 0.768   |
| V13RWB   | 0.003563 | 0.001691 | 2.107000 | 1.922100 | 1.792200 | 0.944 | 0.997   |
| V41SPP   | 0.003386 | 0.002315 | 1.463000 | 1.200500 | 1.440100 | 0.950 | 0.889   |
| V87LFC   | 0.003273 | 0.005878 | 0.557000 | 0.289600 | 0.000000 | 0.955 | 0.554   |
| V45MGLK  | 0.003023 | 0.005428 | 0.557000 | 0.271900 | 0.000000 | 0.961 | 0.783   |
| V8WNHE   | 0.002915 | 0.000912 | 3.194000 | 1.843200 | 1.595400 | 0.966 | 0.940   |
| V53MB    | 0.002779 | 0.004990 | 0.557000 | 0.250000 | 0.000000 | 0.971 | 0.756   |
| V94RBEE  | 0.002642 | 0.004736 | 0.558000 | 0.000000 | 0.256000 | 0.976 | 0.433   |
| V96DF    | 0.002642 | 0.004736 | 0.558000 | 0.000000 | 0.256000 | 0.980 | 0.359   |
| V76GBB   | 0.002548 | 0.004573 | 0.557000 | 0.000000 | 0.210200 | 0.985 | 0.394   |
| V16ST    | 0.002444 | 0.001755 | 1.393000 | 1.493300 | 1.425700 | 0.989 | 0.311   |
| V88WG    | 0.002395 | 0.004296 | 0.557000 | 0.000000 | 0.210200 | 0.993 | 0.403   |
| V95HBC   | 0.002395 | 0.004296 | 0.557000 | 0.000000 | 0.210200 | 0.997 | 0.042 * |
| V24BFCS  | 0.001441 | 0.001298 | 1.110000 | 1.112500 | 1.014400 | 1.000 | 0.983   |
| V20PCU   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |
| V31AMAG  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |
| V32SCC   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |
| V37WHIP  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |
| V47RFC   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |
| V48YTBC  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |
| V49LYRE  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |
| V50CHE   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |
| V510WH   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |
| V52TRM   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |
| V54STHR  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |
| V55LHE   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |
| V57PINK  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |
| V59YR    | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |
| V60LFB   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |
| V62RBTR  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |
| V64BELL  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |
| V67GGC   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |
| V68PTL   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |
| V69SKF   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |
| V71PD0V  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |
| V72CRP   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |
| V73JW    | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |
| V74BCHE  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |
| V77RRP   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |
| V78LLOR  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |
| V80RF    | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |
| V82AZKF  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |
| V83SFC   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |
| V84YRTH  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |
| V85ROSE  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |
| V86BC00  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |
| V89PC00  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |
| V92NFB   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |
| V93DB    | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |
| V97PCL   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |
| V98FLAME | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |
| V99WWT   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |
| V100WBWS | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |
| V101LCOR | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |
| V102KING | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |

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Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Contrast: Gippsland Ma\_River Red Gu

|         | average  | sd       | ratio     | ava      | avb      | cumsum | p         |
|---------|----------|----------|-----------|----------|----------|--------|-----------|
| V32SCC  | 0.025596 | 0.002660 | 9.624000  | 0.000000 | 2.225000 | 0.041  | 0.001 *** |
| V13RWB  | 0.022078 | 0.001581 | 13.960000 | 1.922100 | 0.000000 | 0.077  | 0.001 *** |
| V10WBSW | 0.020296 | 0.002871 | 7.069000  | 1.760400 | 0.000000 | 0.110  | 0.001 *** |
| V27NHHE | 0.018894 | 0.005104 | 3.702000  | 1.646300 | 0.000000 | 0.140  | 0.001 *** |

|          |          |          |           |          |          |       |       |     |
|----------|----------|----------|-----------|----------|----------|-------|-------|-----|
| V2EYR    | 0.018457 | 0.002843 | 6.492000  | 1.599500 | 0.000000 | 0.170 | 0.001 | *** |
| V26WWCH  | 0.018125 | 0.001284 | 14.119000 | 0.000000 | 1.578000 | 0.199 | 0.002 | **  |
| V8WNHE   | 0.018039 | 0.006464 | 2.791000  | 1.843200 | 0.289600 | 0.228 | 0.001 | *** |
| V30BTR   | 0.017484 | 0.001776 | 9.844000  | 0.000000 | 1.520300 | 0.257 | 0.001 | *** |
| V31AMAG  | 0.016036 | 0.001581 | 10.146000 | 0.000000 | 1.393900 | 0.283 | 0.001 | *** |
| V60LFB   | 0.015147 | 0.003696 | 4.098000  | 0.000000 | 1.312300 | 0.307 | 0.001 | *** |
| V38GAL   | 0.014698 | 0.007189 | 2.044000  | 0.323400 | 1.582800 | 0.331 | 0.001 | *** |
| V47RFC   | 0.014352 | 0.001420 | 10.106000 | 0.000000 | 1.251400 | 0.354 | 0.001 | *** |
| V36YFHE  | 0.014260 | 0.009360 | 1.524000  | 1.243700 | 0.000000 | 0.377 | 0.013 | *   |
| V7WEHE   | 0.013756 | 0.008586 | 1.602000  | 1.241500 | 0.000000 | 0.399 | 0.004 | **  |
| V41SPP   | 0.013744 | 0.002390 | 5.750000  | 1.200500 | 0.000000 | 0.421 | 0.003 | **  |
| V69SKF   | 0.013273 | 0.001217 | 10.902000 | 0.000000 | 1.156400 | 0.443 | 0.001 | *** |
| V4BTH    | 0.013108 | 0.008293 | 1.581000  | 1.184200 | 0.000000 | 0.464 | 0.077 | .   |
| V70RSL   | 0.011108 | 0.007818 | 1.421000  | 0.236400 | 1.085200 | 0.482 | 0.010 | **  |
| V45MGLK  | 0.010974 | 0.005854 | 1.875000  | 0.271900 | 1.216900 | 0.500 | 0.002 | **  |
| V98FLAME | 0.010959 | 0.006960 | 1.575000  | 0.000000 | 0.964700 | 0.517 | 0.003 | **  |
| V73JW    | 0.010616 | 0.006635 | 1.600000  | 0.000000 | 0.934100 | 0.534 | 0.001 | *** |
| V34WSW   | 0.010143 | 0.007370 | 1.376000  | 1.856000 | 0.961700 | 0.551 | 0.174 |     |
| V52TRM   | 0.009996 | 0.006422 | 1.556000  | 0.000000 | 0.871300 | 0.567 | 0.009 | **  |
| V18YTH   | 0.009754 | 0.010215 | 0.955000  | 0.000000 | 0.845000 | 0.583 | 0.253 |     |
| V93DB    | 0.009507 | 0.005742 | 1.656000  | 0.000000 | 0.829800 | 0.598 | 0.001 | *** |
| V91NMIN  | 0.008898 | 0.008710 | 1.022000  | 0.719700 | 0.767400 | 0.612 | 0.022 | *   |
| V15STTH  | 0.008597 | 0.008559 | 1.004000  | 0.323400 | 0.745400 | 0.626 | 0.498 |     |
| V36F     | 0.008595 | 0.004954 | 1.735000  | 0.646500 | 1.077300 | 0.640 | 0.116 |     |
| V6WTTR   | 0.008498 | 0.008891 | 0.956000  | 0.000000 | 0.736100 | 0.654 | 0.360 |     |
| V94RBEE  | 0.008405 | 0.005103 | 1.647000  | 0.000000 | 0.732200 | 0.667 | 0.001 | *** |
| V65LWB   | 0.008183 | 0.008484 | 0.965000  | 0.741500 | 0.000000 | 0.681 | 0.136 |     |
| V23RBFT  | 0.008089 | 0.005959 | 1.357000  | 0.937700 | 0.456500 | 0.694 | 0.611 |     |
| V63DWS   | 0.008008 | 0.008299 | 0.965000  | 0.724800 | 0.000000 | 0.707 | 0.061 | .   |
| V28VS    | 0.007914 | 0.008215 | 0.963000  | 0.000000 | 0.686000 | 0.720 | 0.576 |     |
| V35STP   | 0.007822 | 0.007933 | 0.986000  | 0.993200 | 1.638300 | 0.732 | 0.565 |     |
| V56WH    | 0.007810 | 0.005625 | 1.388000  | 0.674800 | 0.765000 | 0.745 | 0.184 |     |
| V21ESP   | 0.007696 | 0.008143 | 0.945000  | 0.644000 | 0.000000 | 0.757 | 0.708 |     |
| V59YR    | 0.007520 | 0.007810 | 0.963000  | 0.000000 | 0.651800 | 0.769 | 0.001 | *** |
| V42SIL   | 0.007277 | 0.009602 | 0.758000  | 0.420400 | 0.442300 | 0.781 | 0.775 |     |
| V56FTC   | 0.007269 | 0.006882 | 1.056000  | 0.615700 | 0.220000 | 0.793 | 0.361 |     |
| V88WG    | 0.007075 | 0.007399 | 0.956000  | 0.000000 | 0.612900 | 0.804 | 0.001 | *** |
| V90WTG   | 0.007075 | 0.007399 | 0.956000  | 0.000000 | 0.612900 | 0.816 | 0.015 | *   |
| V99WWT   | 0.007063 | 0.007361 | 0.959000  | 0.000000 | 0.615300 | 0.827 | 0.001 | *** |
| V580B0   | 0.007005 | 0.007268 | 0.964000  | 0.000000 | 0.611700 | 0.838 | 0.081 | .   |
| V9SFW    | 0.006866 | 0.003914 | 1.754000  | 1.734700 | 1.145500 | 0.849 | 0.360 |     |
| V33RWH   | 0.006745 | 0.006466 | 1.043000  | 0.579800 | 0.532700 | 0.860 | 0.593 |     |
| V20PCU   | 0.006711 | 0.007071 | 0.949000  | 0.000000 | 0.602800 | 0.871 | 0.483 |     |
| V16LR    | 0.006244 | 0.004942 | 1.264000  | 1.184900 | 1.252000 | 0.881 | 0.827 |     |
| V72CRP   | 0.006087 | 0.006321 | 0.963000  | 0.000000 | 0.545600 | 0.891 | 0.001 | *** |
| V92NFB   | 0.006062 | 0.006313 | 0.960000  | 0.000000 | 0.513100 | 0.901 | 0.007 | **  |
| V71PDOV  | 0.006011 | 0.006489 | 0.926000  | 0.000000 | 0.530500 | 0.910 | 0.002 | **  |
| V12LK    | 0.005937 | 0.005887 | 1.009000  | 0.801000 | 0.910300 | 0.920 | 0.064 | .   |
| V14AUR   | 0.005129 | 0.006478 | 0.792000  | 0.962100 | 1.237700 | 0.928 | 0.688 |     |
| V87LFC   | 0.005124 | 0.006716 | 0.763000  | 0.289600 | 0.314400 | 0.937 | 0.223 |     |
| V84YRTH  | 0.004535 | 0.008137 | 0.557000  | 0.000000 | 0.389600 | 0.944 | 0.040 | *   |
| V101LCOR | 0.004040 | 0.007245 | 0.558000  | 0.000000 | 0.370000 | 0.951 | 0.001 | *** |
| V53MB    | 0.004022 | 0.005167 | 0.778000  | 0.250000 | 0.210200 | 0.957 | 0.570 |     |
| V29CST   | 0.003883 | 0.005041 | 0.770000  | 1.059100 | 0.776000 | 0.963 | 0.980 |     |
| V46BHHE  | 0.003418 | 0.006118 | 0.559000  | 0.307900 | 0.000000 | 0.969 | 0.985 |     |
| V74BCHE  | 0.003144 | 0.005642 | 0.557000  | 0.000000 | 0.261700 | 0.974 | 0.002 | **  |
| V81SHBC  | 0.003030 | 0.005425 | 0.559000  | 0.271900 | 0.000000 | 0.979 | 0.988 |     |
| V17WPHE  | 0.002749 | 0.001706 | 1.611000  | 2.247700 | 2.034900 | 0.983 | 0.978 |     |
| V24BFCS  | 0.002646 | 0.002193 | 1.206000  | 1.112500 | 1.328700 | 0.988 | 0.911 |     |
| V95HBC   | 0.002389 | 0.004287 | 0.557000  | 0.000000 | 0.198800 | 0.991 | 0.043 | *   |
| V16ST    | 0.002019 | 0.001634 | 1.236000  | 1.493300 | 1.343300 | 0.995 | 0.593 |     |
| V19ER    | 0.001923 | 0.001172 | 1.640000  | 1.308600 | 1.397300 | 0.998 | 0.995 |     |
| V25WAG   | 0.001422 | 0.001101 | 1.292000  | 1.376000 | 1.365200 | 1.000 | 0.956 |     |
| V11CR    | 0.000000 | 0.000000 | NaN       | 0.000000 | 0.000000 | 1.000 | NA    |     |
| V22SCR   | 0.000000 | 0.000000 | NaN       | 0.000000 | 0.000000 | 1.000 | NA    |     |

|          |          |          |     |          |          |       |    |
|----------|----------|----------|-----|----------|----------|-------|----|
| V37WHIP  | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V39FHE   | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V40BRTH  | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V43GCU   | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V44MUSK  | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V48YTBC  | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V49LYRE  | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V50CHE   | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V51OWH   | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V54STHR  | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V55LHE   | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V57PINK  | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V61SPW   | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V62RBTR  | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V64BELL  | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V66CBW   | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V67GGC   | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V68PIL   | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V75RCR   | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V76GBB   | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V77RRP   | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V78LLOR  | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V79YTHE  | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V80RF    | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V82AZKF  | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V83SFC   | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V85ROSE  | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V86BCO0  | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V89PCO0  | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V96DF    | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V97PCL   | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V100WBWS | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V102KING | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |

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Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Contrast: Montane Fore\_Foothills Wo

|         | average  | sd       | ratio    | ava      | avb      | cumsum | p        |
|---------|----------|----------|----------|----------|----------|--------|----------|
| V40BRTH | 0.014435 | 0.009183 | 1.571800 | 0.000000 | 1.423800 | 0.042  | 0.011 *  |
| V18YTH  | 0.012164 | 0.007560 | 1.609000 | 0.000000 | 1.199300 | 0.077  | 0.064 .  |
| V26WWCH | 0.010745 | 0.008003 | 1.342600 | 0.462400 | 1.210300 | 0.108  | 0.138    |
| V22SCR  | 0.010509 | 0.005505 | 1.909200 | 0.317500 | 1.364500 | 0.138  | 0.032 *  |
| V37WHIP | 0.010114 | 0.006054 | 1.670700 | 1.020400 | 0.000000 | 0.168  | 0.022 *  |
| V48YTBC | 0.009915 | 0.005918 | 1.675400 | 0.958400 | 0.000000 | 0.196  | 0.036 *  |
| V83SFC  | 0.009752 | 0.004877 | 1.999500 | 1.220000 | 0.261700 | 0.224  | 0.028 *  |
| V10WBSW | 0.009395 | 0.007408 | 1.268200 | 1.622600 | 0.773600 | 0.251  | 0.379    |
| V67GGC  | 0.009162 | 0.005646 | 1.622600 | 0.926600 | 0.000000 | 0.278  | 0.007 ** |
| V69SKF  | 0.009034 | 0.005447 | 1.658400 | 0.000000 | 0.890600 | 0.304  | 0.027 *  |
| V53MB   | 0.008754 | 0.005324 | 1.644300 | 0.000000 | 0.851600 | 0.329  | 0.017 *  |
| V33RWH  | 0.008585 | 0.007066 | 1.215100 | 0.753700 | 1.597500 | 0.354  | 0.267    |
| V46BHHE | 0.008556 | 0.006287 | 1.361000 | 0.828800 | 0.931700 | 0.379  | 0.484    |
| V13RWB  | 0.008482 | 0.004846 | 1.750400 | 0.662400 | 1.170600 | 0.403  | 0.698    |
| V49LYRE | 0.007927 | 0.004849 | 1.634700 | 0.803100 | 0.000000 | 0.426  | 0.020 *  |
| V68PIL  | 0.007799 | 0.004781 | 1.631300 | 0.785200 | 0.000000 | 0.449  | 0.022 *  |
| V80RF   | 0.007785 | 0.006023 | 1.292600 | 0.913500 | 0.304500 | 0.471  | 0.122    |
| V54STHR | 0.007684 | 0.004615 | 1.665200 | 0.758900 | 0.000000 | 0.493  | 0.002 ** |
| V23RBFT | 0.007670 | 0.005612 | 1.366800 | 0.883700 | 0.243500 | 0.516  | 0.719    |
| V28VS   | 0.007670 | 0.007611 | 1.007700 | 0.709200 | 1.088900 | 0.538  | 0.664    |
| V43GCU  | 0.007662 | 0.005636 | 1.359400 | 1.256100 | 0.583000 | 0.560  | 0.212    |
| V16LR   | 0.007478 | 0.006026 | 1.241100 | 1.023200 | 1.479800 | 0.582  | 0.657    |
| V20PCU  | 0.006977 | 0.006456 | 1.080800 | 1.268100 | 0.673000 | 0.602  | 0.390    |
| V19ER   | 0.006769 | 0.006493 | 1.042500 | 0.266900 | 0.654600 | 0.621  | 0.733    |
| V29CST  | 0.006698 | 0.006812 | 0.983200 | 0.314400 | 0.659900 | 0.641  | 0.519    |
| V50CHE  | 0.006550 | 0.006045 | 1.083500 | 0.509600 | 0.384600 | 0.659  | 0.446    |

|          |          |          |          |          |          |       |       |
|----------|----------|----------|----------|----------|----------|-------|-------|
| V35STP   | 0.006186 | 0.006202 | 0.997400 | 0.944200 | 1.011200 | 0.677 | 0.855 |
| V42SIL   | 0.006120 | 0.004742 | 1.290500 | 1.363600 | 0.761200 | 0.695 | 0.924 |
| V24BFCS  | 0.005843 | 0.005215 | 1.120300 | 0.496700 | 0.794400 | 0.712 | 0.399 |
| V14AUR   | 0.005747 | 0.006056 | 0.948900 | 0.919800 | 0.987000 | 0.729 | 0.589 |
| V55LHE   | 0.005677 | 0.005865 | 0.967900 | 0.590500 | 0.000000 | 0.745 | 0.246 |
| V36YFHE  | 0.005166 | 0.005585 | 0.925100 | 1.134500 | 1.319700 | 0.760 | 0.968 |
| V9SFW    | 0.005041 | 0.006178 | 0.815900 | 1.525000 | 1.081900 | 0.774 | 0.595 |
| V64BELL  | 0.004931 | 0.008845 | 0.557500 | 0.000000 | 0.492000 | 0.789 | 0.381 |
| V86BC00  | 0.004867 | 0.005040 | 0.965600 | 0.481700 | 0.000000 | 0.803 | 0.103 |
| V81SHBC  | 0.004290 | 0.004697 | 0.913300 | 1.191300 | 0.807100 | 0.815 | 0.958 |
| V62RBTR  | 0.004175 | 0.005389 | 0.774800 | 0.281200 | 0.220000 | 0.827 | 0.194 |
| V21ESP   | 0.003642 | 0.002780 | 1.310000 | 1.502100 | 1.346100 | 0.838 | 0.987 |
| V96DF    | 0.003541 | 0.006353 | 0.557400 | 0.000000 | 0.349000 | 0.848 | 0.226 |
| V4BTH    | 0.003496 | 0.002575 | 1.358000 | 2.294000 | 1.975000 | 0.858 | 0.958 |
| V61SPW   | 0.003310 | 0.005938 | 0.557400 | 0.000000 | 0.326200 | 0.868 | 0.255 |
| V8WNHE   | 0.003301 | 0.002559 | 1.290000 | 1.709100 | 1.582200 | 0.877 | 0.899 |
| V85ROSE  | 0.003237 | 0.005792 | 0.558900 | 0.331700 | 0.000000 | 0.886 | 0.337 |
| V7WEHE   | 0.002775 | 0.001747 | 1.588400 | 1.224500 | 1.318200 | 0.894 | 0.952 |
| V27NHHE  | 0.002698 | 0.004842 | 0.557300 | 0.000000 | 0.256000 | 0.902 | 0.975 |
| V580B0   | 0.002685 | 0.004816 | 0.557400 | 0.000000 | 0.261700 | 0.910 | 0.876 |
| V11CR    | 0.002557 | 0.001689 | 1.513300 | 1.710400 | 1.484400 | 0.917 | 0.996 |
| V63DWS   | 0.002440 | 0.004377 | 0.557500 | 0.000000 | 0.243500 | 0.924 | 0.851 |
| V57PINK  | 0.002440 | 0.004365 | 0.558900 | 0.250000 | 0.000000 | 0.931 | 0.460 |
| V76GBB   | 0.002313 | 0.004139 | 0.558900 | 0.220000 | 0.000000 | 0.938 | 0.590 |
| V66CBW   | 0.002257 | 0.004050 | 0.557400 | 0.000000 | 0.220000 | 0.945 | 0.629 |
| V102KING | 0.002240 | 0.004008 | 0.558900 | 0.236400 | 0.000000 | 0.951 | 0.333 |
| V41SPP   | 0.002181 | 0.001351 | 1.614700 | 1.352200 | 1.526000 | 0.958 | 0.940 |
| V510WH   | 0.002147 | 0.003842 | 0.558900 | 0.220000 | 0.000000 | 0.964 | 0.582 |
| V1GST    | 0.002076 | 0.002289 | 0.906700 | 1.379200 | 1.577800 | 0.970 | 0.513 |
| V2EYR    | 0.002039 | 0.001545 | 1.319900 | 1.440100 | 1.316800 | 0.976 | 0.956 |
| V15STTH  | 0.001903 | 0.001654 | 1.150800 | 1.857500 | 1.788200 | 0.981 | 0.991 |
| V56WH    | 0.001891 | 0.001382 | 1.367800 | 1.486100 | 1.309900 | 0.987 | 0.981 |
| V36F     | 0.001741 | 0.001033 | 1.685600 | 1.724600 | 1.569600 | 0.992 | 0.991 |
| V6WTTR   | 0.001138 | 0.000630 | 1.805600 | 1.567100 | 1.579900 | 0.995 | 0.997 |
| V56FTC   | 0.001005 | 0.000717 | 1.402700 | 1.231300 | 1.244700 | 0.998 | 0.999 |
| V12LK    | 0.000762 | 0.000683 | 1.116400 | 1.324100 | 1.350100 | 1.000 | 0.999 |
| V17WPHE  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V25WAG   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V30BTR   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V31AMAG  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V32SCC   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V34WSW   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V38GAL   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V39FHE   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V44MUSK  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V45MGLK  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V47RFC   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V52TRM   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V59YR    | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V60LFB   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V65LWB   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V70RSL   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V71PD0V  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V72CRP   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V73JW    | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V74BCHE  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V75RCR   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V77RRP   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V78LLOR  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V79YTHE  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V82AZKF  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V84YRTH  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V87LFC   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V88WG    | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V89PC00  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |

|          |          |          |     |          |          |       |    |
|----------|----------|----------|-----|----------|----------|-------|----|
| V90WTG   | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V91NMIN  | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V92NFB   | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V93DB    | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V94RBEE  | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V95HBC   | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V97PCL   | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V98FLAME | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V99WWT   | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V100WBWS | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V101LCOR | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |

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Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Contrast: Montane Fore\_Box-Ironbark

|         | average  | sd       | ratio     | ava      | avb      | cumsum | p         |
|---------|----------|----------|-----------|----------|----------|--------|-----------|
| V18YTH  | 0.019614 | 0.002267 | 8.650000  | 0.000000 | 1.909900 | 0.042  | 0.001 *** |
| V10WBSW | 0.016737 | 0.002289 | 7.312000  | 1.622600 | 0.000000 | 0.079  | 0.008 **  |
| V39FHE  | 0.015508 | 0.003007 | 5.157000  | 0.000000 | 1.532700 | 0.112  | 0.003 **  |
| V21ESP  | 0.015277 | 0.002850 | 5.360000  | 1.502100 | 0.000000 | 0.145  | 0.006 **  |
| V40BRTH | 0.013739 | 0.008267 | 1.662000  | 0.000000 | 1.379300 | 0.175  | 0.022 *   |
| V20PCU  | 0.013073 | 0.001670 | 7.829000  | 1.268100 | 0.000000 | 0.203  | 0.009 **  |
| V26WWCH | 0.012932 | 0.006735 | 1.920000  | 0.462400 | 1.655500 | 0.231  | 0.033 *   |
| V13RWB  | 0.012500 | 0.008069 | 1.549000  | 0.662400 | 1.792200 | 0.258  | 0.073 .   |
| V83SFC  | 0.012497 | 0.001221 | 10.232000 | 1.220000 | 0.000000 | 0.285  | 0.003 **  |
| V44MUSK | 0.011793 | 0.007641 | 1.543000  | 0.000000 | 1.180800 | 0.310  | 0.010 **  |
| V580B0  | 0.010960 | 0.002648 | 4.139000  | 0.000000 | 1.059200 | 0.334  | 0.002 **  |
| V22SCR  | 0.010674 | 0.005558 | 1.920000  | 0.317500 | 1.350800 | 0.357  | 0.029 *   |
| V46BHHE | 0.010527 | 0.009610 | 1.095000  | 0.828800 | 1.798900 | 0.380  | 0.121     |
| V75RCR  | 0.010410 | 0.003002 | 3.468000  | 0.000000 | 1.000900 | 0.402  | 0.001 *** |
| V37WHIP | 0.010161 | 0.006135 | 1.656000  | 1.020400 | 0.000000 | 0.424  | 0.025 *   |
| V56FTC  | 0.010047 | 0.005497 | 1.828000  | 1.231300 | 0.281200 | 0.446  | 0.038 *   |
| V48YTBC | 0.009965 | 0.006005 | 1.659000  | 0.958400 | 0.000000 | 0.468  | 0.027 *   |
| V81SHBC | 0.009699 | 0.005338 | 1.817000  | 1.191300 | 0.281200 | 0.489  | 0.022 *   |
| V67GGC  | 0.009203 | 0.005719 | 1.609000  | 0.926600 | 0.000000 | 0.508  | 0.007 **  |
| V80RF   | 0.009127 | 0.005635 | 1.620000  | 0.913500 | 0.000000 | 0.528  | 0.056 .   |
| V16LR   | 0.008206 | 0.006541 | 1.255000  | 1.023200 | 0.576500 | 0.546  | 0.514     |
| V11CR   | 0.008012 | 0.007131 | 1.124000  | 1.710400 | 0.992900 | 0.563  | 0.759     |
| V49LYRE | 0.007963 | 0.004911 | 1.621000  | 0.803100 | 0.000000 | 0.581  | 0.023 *   |
| V4BTH   | 0.007900 | 0.002729 | 2.895000  | 2.294000 | 1.520900 | 0.598  | 0.721     |
| V33RWH  | 0.007855 | 0.007080 | 1.109000  | 0.753700 | 1.512200 | 0.615  | 0.403     |
| V23RBFT | 0.007852 | 0.005901 | 1.331000  | 0.883700 | 0.320500 | 0.631  | 0.674     |
| V68PIL  | 0.007835 | 0.004844 | 1.617000  | 0.785200 | 0.000000 | 0.648  | 0.023 *   |
| V35STP  | 0.007792 | 0.006378 | 1.222000  | 0.944200 | 0.656700 | 0.665  | 0.584     |
| V54STHR | 0.007721 | 0.004680 | 1.650000  | 0.758900 | 0.000000 | 0.682  | 0.005 **  |
| V28VS   | 0.007362 | 0.006716 | 1.096000  | 0.709200 | 0.971800 | 0.698  | 0.719     |
| V19ER   | 0.007289 | 0.005572 | 1.308000  | 0.266900 | 0.846400 | 0.714  | 0.656     |
| V36YFHE | 0.006921 | 0.005444 | 1.271000  | 1.134500 | 1.383500 | 0.729  | 0.918     |
| V66CBW  | 0.006194 | 0.006437 | 0.962000  | 0.000000 | 0.566600 | 0.742  | 0.020 *   |
| V24BFCS | 0.005995 | 0.005034 | 1.191000  | 0.496700 | 1.014400 | 0.755  | 0.383     |
| V2EYR   | 0.005841 | 0.005810 | 1.005000  | 1.440100 | 0.874900 | 0.768  | 0.604     |
| V90WTG  | 0.005838 | 0.006497 | 0.899000  | 0.000000 | 0.542000 | 0.780  | 0.102     |
| V70RSL  | 0.005780 | 0.006013 | 0.961000  | 0.000000 | 0.557400 | 0.793  | 0.354     |
| V55LHE  | 0.005702 | 0.005915 | 0.964000  | 0.590500 | 0.000000 | 0.805  | 0.259     |
| V61SPW  | 0.005691 | 0.005950 | 0.956000  | 0.000000 | 0.593000 | 0.817  | 0.056 .   |
| V50CHE  | 0.005164 | 0.005411 | 0.954000  | 0.509600 | 0.000000 | 0.829  | 0.659     |
| V86BC00 | 0.004890 | 0.005087 | 0.961000  | 0.481700 | 0.000000 | 0.839  | 0.109     |
| V43GCU  | 0.004840 | 0.005651 | 0.857000  | 1.256100 | 0.818400 | 0.850  | 0.869     |
| V29CST  | 0.004252 | 0.005499 | 0.773000  | 0.314400 | 0.256000 | 0.859  | 0.964     |
| V14AUR  | 0.004172 | 0.005415 | 0.771000  | 0.919800 | 1.211600 | 0.868  | 0.827     |
| V36F    | 0.004016 | 0.002231 | 1.800000  | 1.724600 | 1.330200 | 0.876  | 0.691     |
| V79YTHE | 0.003656 | 0.006555 | 0.558000  | 0.000000 | 0.396100 | 0.884  | 0.206     |
| V30BTR  | 0.003534 | 0.006336 | 0.558000  | 0.000000 | 0.382900 | 0.892  | 0.831     |
| V76GBB  | 0.003414 | 0.004520 | 0.755000  | 0.220000 | 0.210200 | 0.899  | 0.289     |

|          |          |          |          |          |          |       |       |
|----------|----------|----------|----------|----------|----------|-------|-------|
| V56WH    | 0.003383 | 0.002341 | 1.445000 | 1.486100 | 1.171900 | 0.907 | 0.805 |
| V85ROSE  | 0.003252 | 0.005834 | 0.557000 | 0.331700 | 0.000000 | 0.914 | 0.340 |
| V62RBTR  | 0.003172 | 0.005698 | 0.557000 | 0.281200 | 0.000000 | 0.920 | 0.356 |
| V8WNHE   | 0.002763 | 0.002154 | 1.282000 | 1.709100 | 1.595400 | 0.926 | 0.951 |
| V15STTH  | 0.002604 | 0.001970 | 1.321000 | 1.857500 | 1.627000 | 0.932 | 0.961 |
| V57PINK  | 0.002450 | 0.004397 | 0.557000 | 0.250000 | 0.000000 | 0.937 | 0.474 |
| V9SFW    | 0.002418 | 0.002010 | 1.203000 | 1.525000 | 1.472800 | 0.943 | 0.953 |
| V94RBEE  | 0.002363 | 0.004237 | 0.558000 | 0.000000 | 0.256000 | 0.948 | 0.586 |
| V96DF    | 0.002363 | 0.004237 | 0.558000 | 0.000000 | 0.256000 | 0.953 | 0.522 |
| V102KING | 0.002249 | 0.004035 | 0.557000 | 0.236400 | 0.000000 | 0.958 | 0.329 |
| V7WEHE   | 0.002224 | 0.001655 | 1.344000 | 1.224500 | 1.296700 | 0.963 | 0.988 |
| V51OWH   | 0.002157 | 0.003870 | 0.557000 | 0.220000 | 0.000000 | 0.967 | 0.606 |
| V88WG    | 0.002119 | 0.003801 | 0.557000 | 0.000000 | 0.210200 | 0.972 | 0.521 |
| V95HBC   | 0.002119 | 0.003801 | 0.557000 | 0.000000 | 0.210200 | 0.976 | 0.321 |
| V12LK    | 0.002086 | 0.000765 | 2.726000 | 1.324100 | 1.122800 | 0.981 | 0.619 |
| V63DWS   | 0.002031 | 0.003641 | 0.558000 | 0.000000 | 0.220000 | 0.985 | 0.911 |
| V42SIL   | 0.002007 | 0.001772 | 1.133000 | 1.363600 | 1.255000 | 0.990 | 1.000 |
| V16ST    | 0.001927 | 0.000613 | 3.143000 | 1.379200 | 1.425700 | 0.994 | 0.639 |
| V41SPP   | 0.001767 | 0.001227 | 1.439000 | 1.352200 | 1.440100 | 0.998 | 0.965 |
| V6WTR    | 0.001142 | 0.000614 | 1.860000 | 1.567100 | 1.525100 | 1.000 | 0.997 |
| V17WPHE  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V25WAG   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V27NHHE  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V31AMAG  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V32SCC   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V34WSW   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V38GAL   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V45MGLK  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V47RFC   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V52TRM   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V53MB    | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V59YR    | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V60LFB   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V64BELL  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V65LWB   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V69SKF   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V71PD0V  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V72CRP   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V73JW    | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V74BCHE  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V77RRP   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V78LLOR  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V82AZKF  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V84YRTH  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V87LFC   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V89PC00  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V91NMIN  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V92NFB   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V93DB    | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V97PCL   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V98FLAME | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V99WWT   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V100WBWS | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V101LCOR | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |

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Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Contrast: Montane Fore\_River Red Gu

|         | average  | sd       | ratio     | ava      | avb      | cumsum | p         |
|---------|----------|----------|-----------|----------|----------|--------|-----------|
| V4BTH   | 0.023240 | 0.001423 | 16.334000 | 2.294000 | 0.000000 | 0.032  | 0.001 *** |
| V32SCC  | 0.022619 | 0.002319 | 9.752000  | 0.000000 | 2.225000 | 0.062  | 0.002 **  |
| V17WPHE | 0.020648 | 0.001817 | 11.364000 | 0.000000 | 2.034900 | 0.090  | 0.011 *   |
| V11CR   | 0.017360 | 0.001975 | 8.790000  | 1.710400 | 0.000000 | 0.114  | 0.002 **  |
| V10WBSW | 0.016529 | 0.001948 | 8.485000  | 1.622600 | 0.000000 | 0.136  | 0.003 **  |

|          |          |          |           |          |          |       |       |     |
|----------|----------|----------|-----------|----------|----------|-------|-------|-----|
| V38GAL   | 0.016118 | 0.001999 | 8.065000  | 0.000000 | 1.582800 | 0.158 | 0.001 | *** |
| V30BTR   | 0.015451 | 0.001552 | 9.955000  | 0.000000 | 1.520300 | 0.179 | 0.001 | *** |
| V21ESP   | 0.015090 | 0.002641 | 5.713000  | 1.502100 | 0.000000 | 0.200 | 0.003 | **  |
| V2EYR    | 0.014573 | 0.001010 | 14.428000 | 1.440100 | 0.000000 | 0.220 | 0.002 | **  |
| V8WNHE   | 0.014442 | 0.006232 | 2.317000  | 1.709100 | 0.289600 | 0.239 | 0.006 | **  |
| V31AMAG  | 0.014171 | 0.001380 | 10.268000 | 0.000000 | 1.393900 | 0.258 | 0.002 | **  |
| V25WAG   | 0.013859 | 0.001151 | 12.039000 | 0.000000 | 1.365200 | 0.277 | 0.003 | **  |
| V41SPP   | 0.013725 | 0.001684 | 8.149000  | 1.352200 | 0.000000 | 0.296 | 0.004 | **  |
| V60LFB   | 0.013381 | 0.003234 | 4.138000  | 0.000000 | 1.312300 | 0.314 | 0.001 | *** |
| V47RFC   | 0.012687 | 0.001289 | 9.845000  | 0.000000 | 1.251400 | 0.331 | 0.001 | *** |
| V43GCU   | 0.012658 | 0.001321 | 9.581000  | 1.256100 | 0.000000 | 0.348 | 0.003 | **  |
| V7WEHE   | 0.012401 | 0.000859 | 14.431000 | 1.224500 | 0.000000 | 0.365 | 0.010 | **  |
| V26WWCH  | 0.012397 | 0.005637 | 2.199000  | 0.462400 | 1.578000 | 0.382 | 0.066 | .   |
| V83SFC   | 0.012343 | 0.000888 | 13.906000 | 1.220000 | 0.000000 | 0.399 | 0.002 | **  |
| V45MGLK  | 0.012337 | 0.001123 | 10.986000 | 0.000000 | 1.216900 | 0.416 | 0.002 | **  |
| V81SHBC  | 0.012054 | 0.001019 | 11.835000 | 1.191300 | 0.000000 | 0.432 | 0.006 | **  |
| V69SKF   | 0.011732 | 0.001094 | 10.720000 | 0.000000 | 1.156400 | 0.448 | 0.003 | **  |
| V42SIL   | 0.011523 | 0.004695 | 2.455000  | 1.363600 | 0.442300 | 0.464 | 0.037 | *   |
| V19ER    | 0.011415 | 0.005033 | 2.268000  | 0.266900 | 1.397300 | 0.479 | 0.027 | *   |
| V36YFHE  | 0.011410 | 0.006858 | 1.664000  | 1.134500 | 0.000000 | 0.495 | 0.126 |     |
| V15STTH  | 0.011292 | 0.008100 | 1.394000  | 1.857500 | 0.745400 | 0.510 | 0.133 |     |
| V70RSL   | 0.010957 | 0.007153 | 1.532000  | 0.000000 | 1.085200 | 0.525 | 0.006 | **  |
| V56FTC   | 0.010350 | 0.004225 | 2.450000  | 1.231300 | 0.220000 | 0.539 | 0.029 | *   |
| V37WHIP  | 0.010040 | 0.006016 | 1.669000  | 1.020400 | 0.000000 | 0.552 | 0.024 | *   |
| V48YTBC  | 0.009839 | 0.005880 | 1.673000  | 0.958400 | 0.000000 | 0.566 | 0.028 | *   |
| V34WSW   | 0.009813 | 0.006345 | 1.546000  | 0.000000 | 0.961700 | 0.579 | 0.197 |     |
| V98FLAME | 0.009699 | 0.006152 | 1.577000  | 0.000000 | 0.964700 | 0.592 | 0.010 | **  |
| V73JW    | 0.009395 | 0.005864 | 1.602000  | 0.000000 | 0.934100 | 0.605 | 0.003 | **  |
| V67GGC   | 0.009094 | 0.005611 | 1.621000  | 0.926600 | 0.000000 | 0.618 | 0.006 | **  |
| V80RF    | 0.009018 | 0.005525 | 1.632000  | 0.913500 | 0.000000 | 0.630 | 0.054 | .   |
| V52TRM   | 0.008836 | 0.005669 | 1.559000  | 0.000000 | 0.871300 | 0.642 | 0.032 | *   |
| V6WTTT   | 0.008775 | 0.007475 | 1.174000  | 1.567100 | 0.736100 | 0.654 | 0.304 |     |
| V18YTH   | 0.008617 | 0.009023 | 0.955000  | 0.000000 | 0.845000 | 0.665 | 0.462 |     |
| V24BFCS  | 0.008611 | 0.005836 | 1.475000  | 0.496700 | 1.328700 | 0.677 | 0.030 | *   |
| V93DB    | 0.008405 | 0.005073 | 1.657000  | 0.000000 | 0.829800 | 0.689 | 0.003 | **  |
| V91NMN   | 0.007934 | 0.008238 | 0.963000  | 0.000000 | 0.767400 | 0.699 | 0.077 | .   |
| V46BHHE  | 0.007914 | 0.008183 | 0.967000  | 0.828800 | 0.000000 | 0.710 | 0.678 |     |
| V49LYRE  | 0.007869 | 0.004819 | 1.633000  | 0.803100 | 0.000000 | 0.721 | 0.027 | *   |
| V33RWH   | 0.007804 | 0.006307 | 1.237000  | 0.753700 | 0.532700 | 0.731 | 0.430 |     |
| V68PIL   | 0.007742 | 0.004750 | 1.630000  | 0.785200 | 0.000000 | 0.742 | 0.021 | *   |
| V54STHR  | 0.007627 | 0.004585 | 1.663000  | 0.758900 | 0.000000 | 0.752 | 0.003 | **  |
| V20PCU   | 0.007523 | 0.006020 | 1.250000  | 1.268100 | 0.602800 | 0.762 | 0.202 |     |
| V94RBEE  | 0.007429 | 0.004516 | 1.645000  | 0.000000 | 0.732200 | 0.772 | 0.004 | **  |
| V56WH    | 0.007231 | 0.004781 | 1.512000  | 1.486100 | 0.765000 | 0.782 | 0.237 |     |
| V28VS    | 0.007207 | 0.007247 | 0.994000  | 0.709200 | 0.686000 | 0.792 | 0.770 |     |
| V29CST   | 0.007165 | 0.004999 | 1.433000  | 0.314400 | 0.776000 | 0.802 | 0.355 |     |
| V35STP   | 0.007050 | 0.006207 | 1.136000  | 0.944200 | 1.638300 | 0.811 | 0.772 |     |
| V59YR    | 0.006644 | 0.006900 | 0.963000  | 0.000000 | 0.651800 | 0.820 | 0.018 | *   |
| V23RBFT  | 0.006615 | 0.005151 | 1.284000  | 0.883700 | 0.456500 | 0.829 | 0.873 |     |
| V36F     | 0.006534 | 0.001821 | 3.589000  | 1.724600 | 1.077300 | 0.838 | 0.450 |     |
| V13RWB   | 0.006293 | 0.007149 | 0.880000  | 0.662400 | 0.000000 | 0.847 | 0.933 |     |
| V16LR    | 0.006278 | 0.005139 | 1.222000  | 1.023200 | 1.252000 | 0.855 | 0.852 |     |
| V88WG    | 0.006250 | 0.006536 | 0.956000  | 0.000000 | 0.612900 | 0.864 | 0.020 | *   |
| V90WTG   | 0.006250 | 0.006536 | 0.956000  | 0.000000 | 0.612900 | 0.872 | 0.066 | .   |
| V99WWT   | 0.006243 | 0.006502 | 0.960000  | 0.000000 | 0.615300 | 0.881 | 0.017 | *   |
| V580B0   | 0.006193 | 0.006425 | 0.964000  | 0.000000 | 0.611700 | 0.889 | 0.252 |     |
| V55LHE   | 0.005637 | 0.005827 | 0.967000  | 0.590500 | 0.000000 | 0.897 | 0.277 |     |
| V72CRP   | 0.005399 | 0.005608 | 0.963000  | 0.000000 | 0.545600 | 0.904 | 0.018 | *   |
| V92NFB   | 0.005341 | 0.005564 | 0.960000  | 0.000000 | 0.513100 | 0.911 | 0.088 | .   |
| V71PDOV  | 0.005320 | 0.005756 | 0.924000  | 0.000000 | 0.530500 | 0.919 | 0.034 | *   |
| V50CHE   | 0.005101 | 0.005325 | 0.958000  | 0.509600 | 0.000000 | 0.926 | 0.639 |     |
| V86BC00  | 0.004831 | 0.005006 | 0.965000  | 0.481700 | 0.000000 | 0.932 | 0.107 |     |
| V9SFW    | 0.004284 | 0.002149 | 1.993000  | 1.525000 | 1.145500 | 0.938 | 0.684 |     |
| V12LK    | 0.004232 | 0.005626 | 0.752000  | 1.324100 | 0.910300 | 0.944 | 0.375 |     |
| V14AUR   | 0.004098 | 0.005269 | 0.778000  | 0.919800 | 1.237700 | 0.949 | 0.862 |     |



|          |          |          |          |          |          |       |       |
|----------|----------|----------|----------|----------|----------|-------|-------|
| V84YRTH  | 0.004003 | 0.007182 | 0.557000 | 0.000000 | 0.389600 | 0.955 | 0.188 |
| V101LCOR | 0.003592 | 0.006442 | 0.558000 | 0.000000 | 0.370000 | 0.960 | 0.190 |
| V22SCR   | 0.003532 | 0.006323 | 0.559000 | 0.317500 | 0.000000 | 0.964 | 0.885 |
| V85ROSE  | 0.003214 | 0.005752 | 0.559000 | 0.331700 | 0.000000 | 0.969 | 0.381 |
| V62RBTR  | 0.003128 | 0.005600 | 0.559000 | 0.281200 | 0.000000 | 0.973 | 0.393 |
| V87LFC   | 0.003051 | 0.005472 | 0.558000 | 0.000000 | 0.314400 | 0.977 | 0.617 |
| V74BCHE  | 0.002765 | 0.004961 | 0.557000 | 0.000000 | 0.261700 | 0.981 | 0.179 |
| V57PINK  | 0.002422 | 0.004335 | 0.559000 | 0.250000 | 0.000000 | 0.984 | 0.494 |
| V76GBB   | 0.002296 | 0.004109 | 0.559000 | 0.220000 | 0.000000 | 0.987 | 0.562 |
| V102KING | 0.002224 | 0.003981 | 0.559000 | 0.236400 | 0.000000 | 0.990 | 0.367 |
| V53MB    | 0.002160 | 0.003875 | 0.557000 | 0.000000 | 0.210200 | 0.993 | 0.910 |
| V51OWH   | 0.002132 | 0.003815 | 0.559000 | 0.220000 | 0.000000 | 0.996 | 0.624 |
| V95HBC   | 0.002101 | 0.003770 | 0.557000 | 0.000000 | 0.198800 | 0.999 | 0.344 |
| V16ST    | 0.000715 | 0.000313 | 2.283000 | 1.379200 | 1.343300 | 1.000 | 1.000 |
| V27NHHE  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V39FHE   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V40BRTH  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V44MUSK  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V61SPW   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V63DWS   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V64BELL  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V65LWB   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V66CBW   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V75RCR   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V77RRP   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V78LLOR  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V79YTHE  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V82AZKF  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V89PC00  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V96DF    | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V97PCL   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V100WBWS | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |

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Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Contrast: Foothills Wo\_Box-Ironbark

|         | average  | sd       | ratio    | ava      | avb      | cumsum | p         |
|---------|----------|----------|----------|----------|----------|--------|-----------|
| V39FHE  | 0.015833 | 0.002871 | 5.514000 | 0.000000 | 1.532700 | 0.047  | 0.001 *** |
| V21ESP  | 0.014135 | 0.003111 | 4.543000 | 1.346100 | 0.000000 | 0.089  | 0.009 **  |
| V44MUSK | 0.012037 | 0.007747 | 1.554000 | 0.000000 | 1.180800 | 0.125  | 0.007 **  |
| V75RCR  | 0.010634 | 0.002985 | 3.562000 | 0.000000 | 1.000900 | 0.157  | 0.001 *** |
| V56FTC  | 0.010423 | 0.005633 | 1.850000 | 1.244700 | 0.281200 | 0.188  | 0.026 *   |
| V16LR   | 0.009727 | 0.006963 | 1.397000 | 1.479800 | 0.576500 | 0.217  | 0.188     |
| V40BRTH | 0.009298 | 0.009205 | 1.010000 | 1.423800 | 1.379300 | 0.245  | 0.206     |
| V69SKF  | 0.009269 | 0.005598 | 1.656000 | 0.890600 | 0.000000 | 0.273  | 0.024 *   |
| V46BHHE | 0.009105 | 0.006329 | 1.439000 | 0.931700 | 1.798900 | 0.300  | 0.349     |
| V53MB   | 0.008985 | 0.005474 | 1.641000 | 0.851600 | 0.000000 | 0.327  | 0.018 *   |
| V580B0  | 0.008774 | 0.004921 | 1.783000 | 0.261700 | 1.059200 | 0.353  | 0.027 *   |
| V35STP  | 0.008296 | 0.006636 | 1.250000 | 1.011200 | 0.656700 | 0.378  | 0.438     |
| V10WBSW | 0.008277 | 0.008818 | 0.939000 | 0.773600 | 0.000000 | 0.402  | 0.651     |
| V18YTH  | 0.007632 | 0.008144 | 0.937000 | 1.199300 | 1.909900 | 0.425  | 0.593     |
| V81SHBC | 0.007489 | 0.005740 | 1.305000 | 0.807100 | 0.281200 | 0.447  | 0.203     |
| V20PCU  | 0.006999 | 0.007268 | 0.963000 | 0.673000 | 0.000000 | 0.468  | 0.347     |
| V19ER   | 0.006999 | 0.005848 | 1.197000 | 0.654600 | 0.846400 | 0.489  | 0.725     |
| V29CST  | 0.006862 | 0.006603 | 1.039000 | 0.659900 | 0.256000 | 0.510  | 0.449     |
| V43GCU  | 0.006594 | 0.005495 | 1.200000 | 0.583000 | 0.818400 | 0.529  | 0.528     |
| V13RWB  | 0.006454 | 0.003518 | 1.834000 | 1.170600 | 1.792200 | 0.549  | 0.929     |
| V11CR   | 0.006420 | 0.006540 | 0.982000 | 1.484400 | 0.992900 | 0.568  | 0.923     |
| V28VS   | 0.006385 | 0.006620 | 0.964000 | 1.088900 | 0.971800 | 0.587  | 0.885     |
| V61SPW  | 0.006374 | 0.006302 | 1.011000 | 0.326200 | 0.593000 | 0.606  | 0.024 *   |
| V66CBW  | 0.006186 | 0.005875 | 1.053000 | 0.220000 | 0.566600 | 0.624  | 0.031 *   |
| V26WWCH | 0.006051 | 0.007359 | 0.822000 | 1.210300 | 1.655500 | 0.642  | 0.871     |
| V90WTG  | 0.005966 | 0.006614 | 0.902000 | 0.000000 | 0.542000 | 0.660  | 0.088 .   |
| V70RSL  | 0.005903 | 0.006116 | 0.965000 | 0.000000 | 0.557400 | 0.678  | 0.368     |

|         |          |          |          |          |          |       |       |
|---------|----------|----------|----------|----------|----------|-------|-------|
| V9SFW   | 0.005785 | 0.005795 | 0.998000 | 1.081900 | 1.472800 | 0.695 | 0.523 |
| V42SIL  | 0.005356 | 0.004742 | 1.130000 | 0.761200 | 1.255000 | 0.711 | 0.958 |
| V2EYR   | 0.005255 | 0.005672 | 0.927000 | 1.316800 | 0.874900 | 0.727 | 0.768 |
| V64BELL | 0.005057 | 0.009076 | 0.557000 | 0.492000 | 0.000000 | 0.742 | 0.274 |
| V4BTH   | 0.005028 | 0.003157 | 1.593000 | 1.975000 | 1.520900 | 0.757 | 0.859 |
| V96DF   | 0.004848 | 0.006302 | 0.769000 | 0.349000 | 0.256000 | 0.771 | 0.074 |
| V23RBFT | 0.004574 | 0.005837 | 0.784000 | 0.243500 | 0.320500 | 0.785 | 0.980 |
| V36YFHE | 0.004518 | 0.002675 | 1.689000 | 1.319700 | 1.383500 | 0.798 | 0.981 |
| V14AUR  | 0.004445 | 0.005204 | 0.854000 | 0.987000 | 1.211600 | 0.812 | 0.817 |
| V50CHE  | 0.004163 | 0.007474 | 0.557000 | 0.384600 | 0.000000 | 0.824 | 0.783 |
| V79YTHE | 0.003726 | 0.006667 | 0.559000 | 0.000000 | 0.396100 | 0.835 | 0.182 |
| V30BTR  | 0.003602 | 0.006445 | 0.559000 | 0.000000 | 0.382900 | 0.846 | 0.822 |
| V63DWS  | 0.003558 | 0.004682 | 0.760000 | 0.243500 | 0.220000 | 0.857 | 0.675 |
| V24BFCS | 0.003325 | 0.004364 | 0.762000 | 0.794400 | 1.014400 | 0.867 | 0.882 |
| V80RF   | 0.003296 | 0.005917 | 0.557000 | 0.304500 | 0.000000 | 0.876 | 0.885 |
| V7WEHE  | 0.003045 | 0.002323 | 1.311000 | 1.318200 | 1.296700 | 0.885 | 0.925 |
| V16ST   | 0.002847 | 0.001972 | 1.444000 | 1.577800 | 1.425700 | 0.894 | 0.152 |
| V27NHHE | 0.002771 | 0.004976 | 0.557000 | 0.256000 | 0.000000 | 0.902 | 0.964 |
| V83SFC  | 0.002755 | 0.004946 | 0.557000 | 0.261700 | 0.000000 | 0.910 | 0.897 |
| V15STTH | 0.002694 | 0.001759 | 1.532000 | 1.788200 | 1.627000 | 0.918 | 0.958 |
| V36F    | 0.002586 | 0.002274 | 1.137000 | 1.569600 | 1.330200 | 0.926 | 0.957 |
| V94RBEE | 0.002409 | 0.004310 | 0.559000 | 0.000000 | 0.256000 | 0.933 | 0.561 |
| V12LK   | 0.002400 | 0.001272 | 1.887000 | 1.350100 | 1.122800 | 0.941 | 0.525 |
| V62RBTR | 0.002317 | 0.004159 | 0.557000 | 0.220000 | 0.000000 | 0.947 | 0.575 |
| V8WNHE  | 0.002311 | 0.000652 | 3.548000 | 1.582200 | 1.595400 | 0.954 | 0.968 |
| V76GBB  | 0.002287 | 0.004092 | 0.559000 | 0.000000 | 0.210200 | 0.961 | 0.586 |
| V22SCR  | 0.002258 | 0.001532 | 1.474000 | 1.364500 | 1.350800 | 0.968 | 0.963 |
| V56WH   | 0.002206 | 0.001744 | 1.265000 | 1.309900 | 1.171900 | 0.975 | 0.958 |
| V88WG   | 0.002163 | 0.003870 | 0.559000 | 0.000000 | 0.210200 | 0.981 | 0.455 |
| V95HBC  | 0.002163 | 0.003870 | 0.559000 | 0.000000 | 0.210200 | 0.987 | 0.254 |
| V33RWH  | 0.001840 | 0.001210 | 1.520000 | 1.597500 | 1.512200 | 0.993 | 0.991 |
| V41SPP  | 0.001677 | 0.001086 | 1.544000 | 1.526000 | 1.440100 | 0.998 | 0.967 |
| V6WTR   | 0.000709 | 0.000533 | 1.332000 | 1.579900 | 1.525100 | 1.000 | 1.000 |
| V17WPHE | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V25WAG  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V31AMAG | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V32SCC  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V34WSW  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V37WHIP | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V38GAL  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V45MGLK | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V47RFC  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V48YIBC | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V49LYRE | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V51OWH  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V52TRM  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V54STHR | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V55LHE  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V57PINK | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V59YR   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V60LFB  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V65LWB  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V67GGC  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V68PIL  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V71PD0V | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V72CRP  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V73JW   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V74BCHE | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V77RRP  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V78LLOR | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V82AZKF | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V84YRTH | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V85ROSE | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V86BC00 | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V87LFC  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |

|          |          |          |     |          |          |       |    |
|----------|----------|----------|-----|----------|----------|-------|----|
| V89PC00  | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V91NMIN  | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V92NFB   | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V93DB    | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V97PCL   | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V98FLAME | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V99WWT   | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V100WBWS | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V101LCOR | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |
| V102KING | 0.000000 | 0.000000 | NaN | 0.000000 | 0.000000 | 1.000 | NA |

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Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Contrast: Foothills Wo\_River Red Gu

|          | average  | sd       | ratio     | ava      | avb      | cumsum | p         |
|----------|----------|----------|-----------|----------|----------|--------|-----------|
| V32SCC   | 0.023093 | 0.001828 | 12.633000 | 0.000000 | 2.225000 | 0.035  | 0.001 *** |
| V17WPHE  | 0.021080 | 0.001241 | 16.985000 | 0.000000 | 2.034900 | 0.067  | 0.008 **  |
| V4BTH    | 0.020489 | 0.002659 | 7.707000  | 1.975000 | 0.000000 | 0.098  | 0.003 **  |
| V38GAL   | 0.016457 | 0.001736 | 9.477000  | 0.000000 | 1.582800 | 0.123  | 0.001 *** |
| V41SPP   | 0.015828 | 0.001134 | 13.956000 | 1.526000 | 0.000000 | 0.146  | 0.001 *** |
| V30BTR   | 0.015775 | 0.001205 | 13.087000 | 0.000000 | 1.520300 | 0.170  | 0.002 **  |
| V11CR    | 0.015410 | 0.001314 | 11.727000 | 1.484400 | 0.000000 | 0.194  | 0.012 *   |
| V40BRTH  | 0.014624 | 0.009247 | 1.582000  | 1.423800 | 0.000000 | 0.216  | 0.019 *   |
| V31AMAG  | 0.014468 | 0.001047 | 13.821000 | 0.000000 | 1.393900 | 0.238  | 0.003 **  |
| V25WAG   | 0.014149 | 0.000727 | 19.475000 | 0.000000 | 1.365200 | 0.259  | 0.001 *** |
| V22SCR   | 0.014138 | 0.001253 | 11.280000 | 1.364500 | 0.000000 | 0.280  | 0.005 **  |
| V21ESP   | 0.013956 | 0.002919 | 4.781000  | 1.346100 | 0.000000 | 0.301  | 0.004 **  |
| V7WEHE   | 0.013692 | 0.003128 | 4.377000  | 1.318200 | 0.000000 | 0.322  | 0.005 **  |
| V2EYR    | 0.013677 | 0.002086 | 6.556000  | 1.316800 | 0.000000 | 0.343  | 0.001 *** |
| V60LFB   | 0.013662 | 0.003176 | 4.302000  | 0.000000 | 1.312300 | 0.363  | 0.001 *** |
| V36YFHE  | 0.013654 | 0.002185 | 6.250000  | 1.319700 | 0.000000 | 0.384  | 0.024 *   |
| V8WNHE   | 0.013515 | 0.005982 | 2.259000  | 1.582200 | 0.289600 | 0.404  | 0.009 **  |
| V47RFC   | 0.012953 | 0.001004 | 12.896000 | 0.000000 | 1.251400 | 0.424  | 0.001 *** |
| V45MGLK  | 0.012595 | 0.000795 | 15.834000 | 0.000000 | 1.216900 | 0.443  | 0.001 *** |
| V13RWB   | 0.012138 | 0.001836 | 6.609000  | 1.170600 | 0.000000 | 0.461  | 0.076 .   |
| V70RSL   | 0.011186 | 0.007252 | 1.543000  | 0.000000 | 1.085200 | 0.478  | 0.005 **  |
| V15STTH  | 0.011042 | 0.008000 | 1.380000  | 1.788200 | 0.745400 | 0.495  | 0.162     |
| V33RWH   | 0.011004 | 0.005906 | 1.863000  | 1.597500 | 0.532700 | 0.511  | 0.043 *   |
| V56FTC   | 0.010724 | 0.004330 | 2.477000  | 1.244700 | 0.220000 | 0.528  | 0.015 *   |
| V34WSW   | 0.010020 | 0.006434 | 1.557000  | 0.000000 | 0.961700 | 0.543  | 0.161     |
| V98FLAME | 0.009901 | 0.006236 | 1.588000  | 0.000000 | 0.964700 | 0.558  | 0.002 **  |
| V46BHHE  | 0.009696 | 0.006173 | 1.571000  | 0.931700 | 0.000000 | 0.573  | 0.259     |
| V73JW    | 0.009590 | 0.005943 | 1.614000  | 0.000000 | 0.934100 | 0.587  | 0.002 **  |
| V18YTH   | 0.009296 | 0.008040 | 1.156000  | 1.199300 | 0.845000 | 0.601  | 0.377     |
| V52TRM   | 0.009021 | 0.005747 | 1.570000  | 0.000000 | 0.871300 | 0.615  | 0.024 *   |
| V6WTTR   | 0.008899 | 0.007759 | 1.147000  | 1.579900 | 0.736100 | 0.628  | 0.265     |
| V93DB    | 0.008581 | 0.005139 | 1.670000  | 0.000000 | 0.829800 | 0.641  | 0.001 *** |
| V42SIL   | 0.008525 | 0.004802 | 1.776000  | 0.761200 | 0.442300 | 0.654  | 0.479     |
| V81SHBC  | 0.008439 | 0.005218 | 1.617000  | 0.807100 | 0.000000 | 0.667  | 0.071 .   |
| V10WBSW  | 0.008169 | 0.008666 | 0.943000  | 0.773600 | 0.000000 | 0.679  | 0.652     |
| V91NMIN  | 0.008103 | 0.008379 | 0.967000  | 0.000000 | 0.767400 | 0.691  | 0.056 .   |
| V19ER    | 0.007916 | 0.006665 | 1.188000  | 0.654600 | 1.397300 | 0.703  | 0.404     |
| V53MB    | 0.007757 | 0.005181 | 1.497000  | 0.851600 | 0.210200 | 0.715  | 0.045 *   |
| V28VS    | 0.007735 | 0.007384 | 1.048000  | 1.088900 | 0.686000 | 0.727  | 0.632     |
| V94RBEE  | 0.007585 | 0.004574 | 1.658000  | 0.000000 | 0.732200 | 0.738  | 0.004 **  |
| V20PCU   | 0.007072 | 0.006526 | 1.084000  | 0.673000 | 0.602800 | 0.749  | 0.311     |
| V29CST   | 0.006866 | 0.005049 | 1.360000  | 0.659900 | 0.776000 | 0.759  | 0.433     |
| V59YR    | 0.006784 | 0.007017 | 0.967000  | 0.000000 | 0.651800 | 0.769  | 0.009 **  |
| V35STP   | 0.006652 | 0.006625 | 1.004000  | 1.011200 | 1.638300 | 0.779  | 0.803     |
| V88WG    | 0.006382 | 0.006647 | 0.960000  | 0.000000 | 0.612900 | 0.789  | 0.010 **  |
| V90WTG   | 0.006382 | 0.006647 | 0.960000  | 0.000000 | 0.612900 | 0.799  | 0.057 .   |
| V99WWT   | 0.006373 | 0.006612 | 0.964000  | 0.000000 | 0.615300 | 0.808  | 0.010 **  |
| V580B0   | 0.006330 | 0.006124 | 1.034000  | 0.261700 | 0.611700 | 0.818  | 0.209     |
| V26WWCH  | 0.005950 | 0.006536 | 0.910000  | 1.210300 | 1.578000 | 0.827  | 0.892     |

|          |          |          |          |          |          |       |         |
|----------|----------|----------|----------|----------|----------|-------|---------|
| V43GCU   | 0.005949 | 0.006303 | 0.944000 | 0.583000 | 0.000000 | 0.836 | 0.743   |
| V9SFW    | 0.005874 | 0.004177 | 1.406000 | 1.081900 | 1.145500 | 0.845 | 0.493   |
| V5GWH    | 0.005644 | 0.004760 | 1.186000 | 1.309900 | 0.765000 | 0.853 | 0.564   |
| V72CRP   | 0.005509 | 0.005702 | 0.966000 | 0.000000 | 0.545600 | 0.862 | 0.012 * |
| V24BFCS  | 0.005494 | 0.005150 | 1.067000 | 0.794400 | 1.328700 | 0.870 | 0.473   |
| V92NFB   | 0.005455 | 0.005659 | 0.964000 | 0.000000 | 0.513100 | 0.878 | 0.081 . |
| V71PD0V  | 0.005431 | 0.005852 | 0.928000 | 0.000000 | 0.530500 | 0.886 | 0.026 * |
| V3GF     | 0.005121 | 0.001989 | 2.575000 | 1.569600 | 1.077300 | 0.894 | 0.576   |
| V64BELL  | 0.004995 | 0.008940 | 0.559000 | 0.492000 | 0.000000 | 0.902 | 0.340   |
| V23RBFT  | 0.004889 | 0.004924 | 0.993000 | 0.243500 | 0.456500 | 0.909 | 0.981   |
| V12LK    | 0.004650 | 0.005759 | 0.808000 | 1.350100 | 0.910300 | 0.916 | 0.271   |
| V14AUR   | 0.004157 | 0.005215 | 0.797000 | 0.987000 | 1.237700 | 0.922 | 0.851   |
| V50CHE   | 0.004108 | 0.007354 | 0.559000 | 0.384600 | 0.000000 | 0.928 | 0.782   |
| V84YRTH  | 0.004088 | 0.007314 | 0.559000 | 0.000000 | 0.389600 | 0.935 | 0.139   |
| V69SKF   | 0.003666 | 0.005226 | 0.701000 | 0.890600 | 1.156400 | 0.940 | 0.816   |
| V101LCOR | 0.003664 | 0.006556 | 0.559000 | 0.000000 | 0.370000 | 0.946 | 0.135   |
| V96DF    | 0.003588 | 0.006422 | 0.559000 | 0.349000 | 0.000000 | 0.951 | 0.223   |
| V61SPW   | 0.003353 | 0.006002 | 0.559000 | 0.326200 | 0.000000 | 0.956 | 0.238   |
| V80RF    | 0.003252 | 0.005822 | 0.559000 | 0.304500 | 0.000000 | 0.961 | 0.907   |
| V87LFC   | 0.003113 | 0.005569 | 0.559000 | 0.000000 | 0.314400 | 0.966 | 0.587   |
| V74BCHE  | 0.002825 | 0.005054 | 0.559000 | 0.000000 | 0.261700 | 0.970 | 0.125   |
| V27NHHE  | 0.002735 | 0.004896 | 0.559000 | 0.256000 | 0.000000 | 0.974 | 0.967   |
| V83SFC   | 0.002720 | 0.004869 | 0.559000 | 0.261700 | 0.000000 | 0.978 | 0.930   |
| V16LR    | 0.002626 | 0.002077 | 1.264000 | 1.479800 | 1.252000 | 0.982 | 0.987   |
| V16ST    | 0.002484 | 0.002388 | 1.040000 | 1.577800 | 1.343300 | 0.986 | 0.288   |
| V63DWS   | 0.002472 | 0.004425 | 0.559000 | 0.243500 | 0.000000 | 0.990 | 0.814   |
| V62RBTR  | 0.002287 | 0.004094 | 0.559000 | 0.220000 | 0.000000 | 0.993 | 0.612   |
| V66CBW   | 0.002287 | 0.004094 | 0.559000 | 0.220000 | 0.000000 | 0.997 | 0.602   |
| V95HBC   | 0.002146 | 0.003840 | 0.559000 | 0.000000 | 0.198800 | 1.000 | 0.244   |
| V37WHIP  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |
| V39FHE   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |
| V44MUSK  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |
| V48YBTC  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |
| V49LYRE  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |
| V51OWH   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |
| V54STHR  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |
| V55LHE   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |
| V57PINK  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |
| V65LWB   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |
| V67GGC   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |
| V68PIL   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |
| V75RCR   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |
| V76GBB   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |
| V77RRP   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |
| V78LLOR  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |
| V79YTHE  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |
| V82AZKF  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |
| V85ROSE  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |
| V86BC00  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |
| V89PC00  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |
| V97PCL   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |
| V100WBWS | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |
| V102KING | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA      |

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Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Contrast: Box-Ironbark\_River Red Gu

|         | average  | sd       | ratio     | ava      | avb      | cumsum | p         |
|---------|----------|----------|-----------|----------|----------|--------|-----------|
| V32SCC  | 0.023209 | 0.002485 | 9.340000  | 0.000000 | 2.225000 | 0.036  | 0.001 *** |
| V17WPHE | 0.021185 | 0.001961 | 10.802000 | 0.000000 | 2.034900 | 0.068  | 0.004 **  |
| V46BHHE | 0.018779 | 0.002092 | 8.977000  | 1.798900 | 0.000000 | 0.097  | 0.001 *** |
| V13RWB  | 0.018603 | 0.002860 | 6.505000  | 1.792200 | 0.000000 | 0.126  | 0.001 *** |
| V38GAL  | 0.016539 | 0.002119 | 7.804000  | 0.000000 | 1.582800 | 0.151  | 0.001 *** |
| V4BTH   | 0.015839 | 0.002532 | 6.255000  | 1.520900 | 0.000000 | 0.175  | 0.013 *   |

|          |          |          |           |          |          |       |       |     |
|----------|----------|----------|-----------|----------|----------|-------|-------|-----|
| V39FHE   | 0.015712 | 0.002885 | 5.447000  | 1.532700 | 0.000000 | 0.200 | 0.004 | **  |
| V41SPP   | 0.014960 | 0.001224 | 12.222000 | 1.440100 | 0.000000 | 0.222 | 0.002 | **  |
| V31AMAG  | 0.014541 | 0.001484 | 9.795000  | 0.000000 | 1.393900 | 0.245 | 0.003 | **  |
| V36YFHE  | 0.014477 | 0.004922 | 2.941000  | 1.383500 | 0.000000 | 0.267 | 0.015 | *   |
| V25WAG   | 0.014220 | 0.001253 | 11.353000 | 0.000000 | 1.365200 | 0.289 | 0.003 | **  |
| V22SCR   | 0.014102 | 0.002784 | 5.066000  | 1.350800 | 0.000000 | 0.311 | 0.005 | **  |
| V40BRTH  | 0.013917 | 0.008320 | 1.673000  | 1.379300 | 0.000000 | 0.332 | 0.012 | *   |
| V8WNHE   | 0.013741 | 0.005715 | 2.405000  | 1.595400 | 0.289600 | 0.353 | 0.006 | **  |
| V60LFB   | 0.013731 | 0.003352 | 4.097000  | 0.000000 | 1.312300 | 0.374 | 0.001 | *** |
| V7WEHE   | 0.013625 | 0.003185 | 4.278000  | 1.296700 | 0.000000 | 0.395 | 0.003 | **  |
| V47RFC   | 0.013017 | 0.001371 | 9.494000  | 0.000000 | 1.251400 | 0.415 | 0.001 | *** |
| V45MGLK  | 0.012658 | 0.001205 | 10.507000 | 0.000000 | 1.216900 | 0.435 | 0.001 | *** |
| V30BTR   | 0.012531 | 0.007039 | 1.780000  | 0.382900 | 1.520300 | 0.454 | 0.011 | *   |
| V69SKF   | 0.012038 | 0.001175 | 10.245000 | 0.000000 | 1.156400 | 0.472 | 0.004 | **  |
| V44MUSK  | 0.011946 | 0.007696 | 1.552000  | 1.180800 | 0.000000 | 0.491 | 0.011 | *   |
| V18YTH   | 0.011253 | 0.009055 | 1.243000  | 1.909900 | 0.845000 | 0.508 | 0.093 | .   |
| V42SIL   | 0.011190 | 0.003985 | 2.808000  | 1.255000 | 0.442300 | 0.525 | 0.065 | .   |
| V35STP   | 0.010880 | 0.007777 | 1.399000  | 0.656700 | 1.638300 | 0.542 | 0.074 | .   |
| V75RCR   | 0.010550 | 0.002971 | 3.552000  | 1.000900 | 0.000000 | 0.558 | 0.001 | *** |
| V33RWH   | 0.010203 | 0.005935 | 1.719000  | 1.512200 | 0.532700 | 0.574 | 0.060 | .   |
| V34WSW   | 0.010070 | 0.006524 | 1.544000  | 0.000000 | 0.961700 | 0.589 | 0.165 | .   |
| V11CR    | 0.009963 | 0.006247 | 1.595000  | 0.992900 | 0.000000 | 0.605 | 0.362 | .   |
| V98FLAME | 0.009949 | 0.006321 | 1.574000  | 0.000000 | 0.964700 | 0.620 | 0.007 | **  |
| V73JW    | 0.009637 | 0.006026 | 1.599000  | 0.000000 | 0.934100 | 0.635 | 0.001 | *** |
| V15STTH  | 0.009594 | 0.007905 | 1.214000  | 1.627000 | 0.745400 | 0.649 | 0.351 | .   |
| V52TRM   | 0.009066 | 0.005827 | 1.556000  | 0.000000 | 0.871300 | 0.663 | 0.021 | *   |
| V2EYR    | 0.009033 | 0.005592 | 1.615000  | 0.874900 | 0.000000 | 0.677 | 0.118 | .   |
| V93DB    | 0.008624 | 0.005214 | 1.654000  | 0.000000 | 0.829800 | 0.690 | 0.001 | *** |
| V70RSL   | 0.008584 | 0.006538 | 1.313000  | 0.557400 | 1.085200 | 0.704 | 0.055 | .   |
| V6WTTR   | 0.008576 | 0.007595 | 1.129000  | 1.525100 | 0.736100 | 0.717 | 0.326 | .   |
| V43GCU   | 0.008275 | 0.004993 | 1.657000  | 0.818400 | 0.000000 | 0.730 | 0.139 | .   |
| V91NMIN  | 0.008145 | 0.008464 | 0.962000  | 0.000000 | 0.767400 | 0.742 | 0.045 | *   |
| V16LR    | 0.007429 | 0.006376 | 1.165000  | 0.576500 | 1.252000 | 0.753 | 0.666 | .   |
| V28VS    | 0.007148 | 0.006803 | 1.051000  | 0.971800 | 0.686000 | 0.764 | 0.793 | .   |
| V29CST   | 0.006978 | 0.005342 | 1.306000  | 0.256000 | 0.776000 | 0.775 | 0.420 | .   |
| V59YR    | 0.006818 | 0.007086 | 0.962000  | 0.000000 | 0.651800 | 0.786 | 0.013 | *   |
| V580B0   | 0.006742 | 0.004954 | 1.361000  | 1.059200 | 0.611700 | 0.796 | 0.157 | .   |
| V90WTG   | 0.006683 | 0.006034 | 1.107000  | 0.542000 | 0.612900 | 0.806 | 0.045 | *   |
| V94RBEE  | 0.006671 | 0.005063 | 1.318000  | 0.256000 | 0.732200 | 0.817 | 0.006 | **  |
| V88WG    | 0.006408 | 0.005997 | 1.069000  | 0.210200 | 0.612900 | 0.826 | 0.014 | *   |
| V99WWT   | 0.006405 | 0.006678 | 0.959000  | 0.000000 | 0.615300 | 0.836 | 0.014 | *   |
| V66CBW   | 0.006280 | 0.006500 | 0.966000  | 0.566600 | 0.000000 | 0.846 | 0.033 | *   |
| V19ER    | 0.006192 | 0.005991 | 1.034000  | 0.846400 | 1.397300 | 0.855 | 0.854 | .   |
| V20PCU   | 0.006104 | 0.006440 | 0.948000  | 0.000000 | 0.602800 | 0.865 | 0.710 | .   |
| V61SPW   | 0.005762 | 0.006007 | 0.959000  | 0.593000 | 0.000000 | 0.874 | 0.055 | .   |
| V23RBFT  | 0.005699 | 0.005229 | 1.090000  | 0.320500 | 0.456500 | 0.882 | 0.967 | .   |
| V72CRP   | 0.005535 | 0.005754 | 0.962000  | 0.000000 | 0.545600 | 0.891 | 0.012 | *   |
| V92NFB   | 0.005483 | 0.005717 | 0.959000  | 0.000000 | 0.513100 | 0.899 | 0.060 | .   |
| V71PD0V  | 0.005457 | 0.005906 | 0.924000  | 0.000000 | 0.530500 | 0.908 | 0.022 | *   |
| V56WH    | 0.004967 | 0.004736 | 1.049000  | 1.171900 | 0.765000 | 0.915 | 0.660 | .   |
| V9SFW    | 0.004265 | 0.002744 | 1.554000  | 1.472800 | 1.145500 | 0.922 | 0.694 | .   |
| V84YRTH  | 0.004108 | 0.007375 | 0.557000  | 0.000000 | 0.389600 | 0.928 | 0.151 | .   |
| V56FTC   | 0.003830 | 0.004875 | 0.786000  | 0.281200 | 0.220000 | 0.934 | 0.965 | .   |
| V12LK    | 0.003782 | 0.004861 | 0.778000  | 1.122800 | 0.910300 | 0.940 | 0.383 | .   |
| V79YTHE  | 0.003700 | 0.006623 | 0.559000  | 0.396100 | 0.000000 | 0.946 | 0.175 | .   |
| V101LCOR | 0.003681 | 0.006605 | 0.557000  | 0.000000 | 0.370000 | 0.951 | 0.135 | .   |
| V24BFCS  | 0.003288 | 0.002264 | 1.452000  | 1.014400 | 1.328700 | 0.956 | 0.900 | .   |
| V95HBC   | 0.003249 | 0.004283 | 0.759000  | 0.210200 | 0.198800 | 0.961 | 0.030 | *   |
| V36F     | 0.003169 | 0.001979 | 1.601000  | 1.330200 | 1.077300 | 0.966 | 0.815 | .   |
| V87LFC   | 0.003127 | 0.005611 | 0.557000  | 0.000000 | 0.314400 | 0.971 | 0.616 | .   |
| V74BCHE  | 0.002840 | 0.005099 | 0.557000  | 0.000000 | 0.261700 | 0.975 | 0.111 | .   |
| V81SHBC  | 0.002627 | 0.004702 | 0.559000  | 0.281200 | 0.000000 | 0.979 | 0.995 | .   |
| V96DF    | 0.002392 | 0.004281 | 0.559000  | 0.256000 | 0.000000 | 0.983 | 0.455 | .   |
| V76GBB   | 0.002269 | 0.004061 | 0.559000  | 0.210200 | 0.000000 | 0.986 | 0.594 | .   |
| V53MB    | 0.002217 | 0.003979 | 0.557000  | 0.000000 | 0.210200 | 0.990 | 0.863 | .   |

|          |          |          |          |          |          |       |       |
|----------|----------|----------|----------|----------|----------|-------|-------|
| V63DWS   | 0.002056 | 0.003679 | 0.559000 | 0.220000 | 0.000000 | 0.993 | 0.910 |
| V16ST    | 0.001950 | 0.001137 | 1.714000 | 1.425700 | 1.343300 | 0.996 | 0.650 |
| V26WWCH  | 0.001348 | 0.001039 | 1.297000 | 1.655500 | 1.578000 | 0.998 | 0.983 |
| V14AUR   | 0.001268 | 0.000601 | 2.111000 | 1.211600 | 1.237700 | 1.000 | 0.988 |
| V10WBSW  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V21ESP   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V27NHHE  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V37WHIP  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V48YTBC  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V49LYRE  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V50CHE   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V51OWH   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V54STHR  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V55LHE   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V57PINK  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V62RBTR  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V64BELL  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V65LWB   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V67GGC   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V68PIL   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V77RRP   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V78LLOR  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V80RF    | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V82AZKF  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V83SFC   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V85ROSE  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V86BC00  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V89PC00  | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V97PCL   | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V100WBWS | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |
| V102KING | 0.000000 | 0.000000 | NaN      | 0.000000 | 0.000000 | 1.000 | NA    |

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Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Permutation: free

Number of permutations: 999

The SIMPER analysis shows which species contribute most to the dissimilarity between habitat groups. This helps identify indicator species or species that drive community differences.